BEGINNING WITH JOOMLA!® CMS

web designing with Joomla!™ for beginners

...because open source matters

by Utsav Bhanja

Beginning with Joomla! CMS

WEB DESIGNING WITH JOOMLA! ® FOR BEGINNERS

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Maa, this is for you…

Beginning with Joomla! CMS

WEB DESIGNING WITH JOOMLA! FOR BEGINNERS

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References

docs.joomla.org

Joomla! Open Source CMS documentation

w3schools.com

INTRODUCTION

The art of website designing involves a set of tools that each web designer chooses to achieve the final goal, i.e., creating a fully functional website or web-software. These tools comprise of programming languages, editors, deployment tools and elementary tools. However, the term ‘design’ is in itself a vast thing to explore or achieve excellence in. It is critically understood that everyone can be a designer, however to be a good designer it takes not only the technical knowledge but also the keen insight to experiment with engineering tact, color combination, size & styling of web elements, presentation of site data and above all, performance of the final product. Here performance is referred to speed, interactivity and consistency of presentation. For e.g. Mr. A created a wonderful bright website with a lot of interactive flash content loaded in it, owing to which the website opens up slowly in most computers and hardly opens in some mobile devices. On the other hand, Mr. B created a different website which is target oriented with focused data, lesser visual elements and a few pictorial sober elements. His website opens up pretty fast in almost all devices and browsers but lacks the bright visual interactivity which Mr. A’s website possesses. In this case, Mr. B’s website would be of more value than that of Mr. A, because in today’s world, no one has got the time to sit and wait. Everyone needs a faster access to solution and that is what should be the primary target of the web designer – to deliver a well-engineered website in its true sense that includes small visual elements, interactive functions, proper positioning of data, integrity of the core system and flexibility of design layout and clarity of navigation.

Initially the set of tools we have talked about differ from designer to designer. In this book we focus on content management systems – a vital tool of designing the web in modern world, with specialization on Joomla! ®.

The main purpose of this book is to make you understand the **basics of Joomla! CMS**. By the time you have finished reading this book, you will know what Joomla is, how to install it & how to deploy a simple website with it. Primarily meant for beginners, this book will give you a clear picture of the tools that can help you become a professional in future.

Let’s begin with Joomla!

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1.0 BASIC CONCEPTS

As I have stated earlier, I would focus mainly on two things – Design and Commercial Aspect. But to get started with your new web designing business, you need to understand certain basic things.

1.1 What is HTML?

HTML or **H**yper **T**ext **M**arkup **L**anguage is a language that describes a web page. It consists of markup tags known as HTML Tags. These Markup Tags are defined with angle brackets, for e.g., <html> where 'html' is a keyword enclosed within the '<>'. Mostly tags are defined in pairs for allocating the functional scope of a particular keyword, for e.g., <h1> and </h1>.

1.1.2 Example for HTML

We use a Notepad to write the following code and save it as **example.html** or **example.htm** file –

```
<!DOCTYPE html>
<html>
<head>
<title>My Site.com</title>
</head>
<body>
<h1>My Site's Heading</h1>
<p>Welcome to My Site</p>

</body>
</html>
```

The **OUTPUT** of the above section of HTML code –

Although I won't go into details of HTML coding, but I will surely let you know that introduction of various smart devices like smart phones, iPads, internet TV sets, etc. has challenged web designers over the years to get the perfect adaptable layout. This has led to the need and development of newer versions of HTML that can direct the web page to adapt to the output screen size automatically. Newer version like HTML5 is widely implemented to achieve such goals.

You can learn HTML in detail from the following link –

http://www.w3schools.com/html/html_intro.asp

1.1.3 HTML Tags

Here is a list of commonly used **HTML / HTML5 Tags**:

Tags are sorted in alphabetical order.

Tags	Description
<!DOCTYPE>	Defines the document type
<a>	Defines a hyperlink
<abbr>	Defines an abbreviation
<address>	Defines contact information for the author/owner of a document
<article>	Defines an article
<b>	Defines bold text
<base>	Specifies the base URL/target for all relative URLs in a document
<blockquote>	Defines a section that is quoted from another source
<body>	Defines the document's body
* *	Defines a single line break
<button>	Defines a clickable button
<canvas>	Used to draw graphics, on the fly, via scripting (usually JavaScript)
<caption>	Defines a table caption
<cite>	Defines the title of a work
<code>	Defines a piece of computer code
<col>	Specifies column properties for each column within a <colgroup> element
<colgroup>	Specifies a group of one or more columns in a table for formatting

<datalist>	Specifies a list of pre-defined options for input controls
<details>	Defines additional details that the user can view or hide
<dialog>	Defines a dialog box or window
	Defines a section in a document
	Defines emphasized text
<embed>	Defines a container for an external (non-HTML) application
<fieldset>	Groups related elements in a form
<footer>	Defines a footer for a document or section
<form>	Defines an HTML form for user input
<h1> to <h6>	Defines HTML headings
<head>	Defines information about the document
<header>	Defines a header for a document or section
<html>	Defines the root of an HTML document
<i>	Defines a part of text in an alternate voice or mood
<iframe>	Defines an inline frame
<img>	Defines an image
<input>	Defines an input control
<label>	Defines a label for an <input> element
<legend>	Defines a caption for a <fieldset> element
<li>	Defines a list item
<menu>	Defines a list/menu of commands
<menuitem>	Defines a command/menu item that the user can invoke from a popup menu
<meta>	Defines metadata about an HTML document
<nav>	Defines navigation links
<object>	Defines an embedded object
<ol>	Defines an ordered list

<optgroup>	Defines a group of related options in a drop-down list
<option>	Defines an option in a drop-down list
<output>	Defines the result of a calculation
<p>	Defines a paragraph
<param>	Defines a parameter for an object
<pre>	Defines preformatted text
<q>	Defines a short quotation
<ruby>	Defines a ruby annotation (for East Asian typography)
<script>	Defines a client-side script
<select>	Defines a drop-down list
<small>	Defines smaller text
	Defines a section in a document
	Defines important text
<style>	Defines style information for a document
<table>	Defines a table
<tbody>	Groups the body content in a table
<td>	Defines a cell in a table
<textarea>	Defines a multiline input control (text area)
<tfoot>	Groups the footer content in a table
<th>	Defines a header cell in a table
<thead>	Groups the header content in a table
<time>	Defines a date/time
<title>	Defines a title for the document
<tr>	Defines a row in a table
<u>	Defines text that should be stylistically different from normal text
<ul>	Defines an unordered list

<var>	Defines a variable
<video>	Defines a video or movie
<wbr>	Defines a possible line-break

The Tags listed above can be used as a reference while you code your HTML program. Although in general we do not require more than this for a simple or medium site, you can always refer to the following link for a complete list of HTML Tags and understanding –

http://www.w3schools.com/tags/

1.2 CSS Overview

A web page can be styled with use of CSS – **C**ascading **S**tyle **S**heet. A CSS document is a separate file that stores styling parameters and values of an HTML page from font family to size, color and text transformation, etc.

HTML was never created to declare formatting tags but to define the content of a document, like <h1> and <p> tags. When HTML tags like <font> and other styling attributes were added to the HTML 3.2 specification, development of large and complex websites became a long and costly method. To solve this problem, the World Wide Web Consortium (W3C) created **CSS**.

1.2.1 Structure of CSS

A CSS consists of two blocks – **Selector** and **Declaration**.

A Selector refers to the HTML Tag name for e.g., <h1> and Declaration includes the styling parameters for the particular Selector, for e.g., { font-size: 16px } where 'font-size' is the property and '16px' is the value of property.

A CSS file type ends with a '.css' file extension.

1.2.2 Example of CSS

Following is an example of how CSS is written in a sample file: **mysite.css –**

```
h1 {color: cyan;}
p {margin-left: 10px;}
body {background-image: url("images/mysite-bg.png");}
```

1.2.3 Inserting CSS into HTML

Now we will insert **mysite.css** into an HTML document by External Style Sheet referencing –

The <link> tag is used to place the link to the CSS file we are inserting, after the <head> tag of the HTML document.

```
<head>
<link rel="stylesheet" type="text/css" href="mysite.css">
</head>
```

There are of course other ways to insert CSS into an HTML document, the details of which you can find out from the following link along with specific CSS tutorials –

http://www.w3schools.com/css/css_syntax.asp

I will proceed to the topic of concentration – Joomla! in the next chapter.

2.0 UNDERSTANDING JOOMLA!

Joomla is a **Content Management System** built with **PHP** and **MySQL**. It enables you to create almost any type of website starting from a simple portfolio site to a complex social site. Big names like Ebay, Harvard University Graduate School of Arts & Sciences, Linux, etc. use Joomla for building their web presence as reported in the Joomla Magazine 2012 July issue. But before going into the details of Joomla, let's understand what a content management system actually is.

2.1 Content Management System

A content management system is a software that helps you create and manage a website. It provides you with all the important tools to build a website in an organized visual manner. There are several content management systems out there, with Joomla being the loved one by developers and designers for creating websites of various sizes. As Joomla documentation states, it separates the creation of your contents from the mechanics required to present it on the web.

While a website appears as a presentation of all contents along with its look and feel, a CMS methodically differentiates these two aspects of a website. While all the data is saved in a database, the styling is organized in the website template. A CMS combines both the template with the database to display the website the way it is.

2.1.2 PHP Overview

PHP is a scripting language. It lets you interact with your web pages. PHP enables you to create complex loops and make comparisons. You can make your web pages more dynamic with added functions to generate specialized data out of various complicated calculations and Boolean choices.

The best part of PHP is, it beautifully embeds into an existing HTML to add more functions to the page and not making HTML an obsolete technology. While HTML can be used to design web pages, PHP combined with a MySQL database gives it the power and resource to collect, process, store and retrieve data thereby interacting with the user.

2.1.3 MySQL Database Overview

SQL stands for **"Structured Query Language"** which is the standard language used to interact with databases. MySQL is a relational database system, used to store large amount of information. MySQL when coupled with PHP, functions with ease just like it does in case of Joomla, which is also a live example of PHP and MySQL combination. MySQL can store various file types of different sizes in a tabular format. Generally, a server side MySQL database can be accessed with a MyPHP Admin.

You will learn about accessing and modifying database tables in our next book on applications of Joomla!.

2.2 Ends of a Joomla Website

A Joomla website is basically a combination of two sites – Frontend and Backend.

FRONTEND is the public output. It is the site which appears to a visitor on opening your website.

BACKEND is the Joomla administration panel for you to create content and design your website.

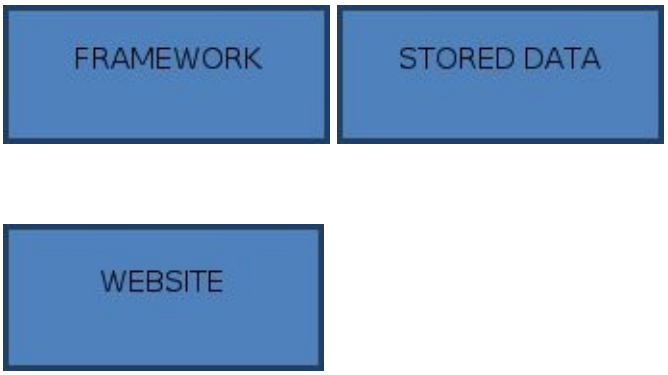
These two separate sites are linked and any change you commit in the Backend, reflects on the Frontend. However, technically a Joomla Backend is also like a visually understandable fully functional site, you might find it easier to relate to as a Control Panel of your computer, only with several options to play around with your website's functionality, content and looks.

A Joomla Backend can be accessed at: example.com/**administrator**

It is essential to understand that the Frontend and Backend of a Joomla installation is just a file based framework and every bit of content, parameters and respective values are actually stored in the MySQL Database tables.

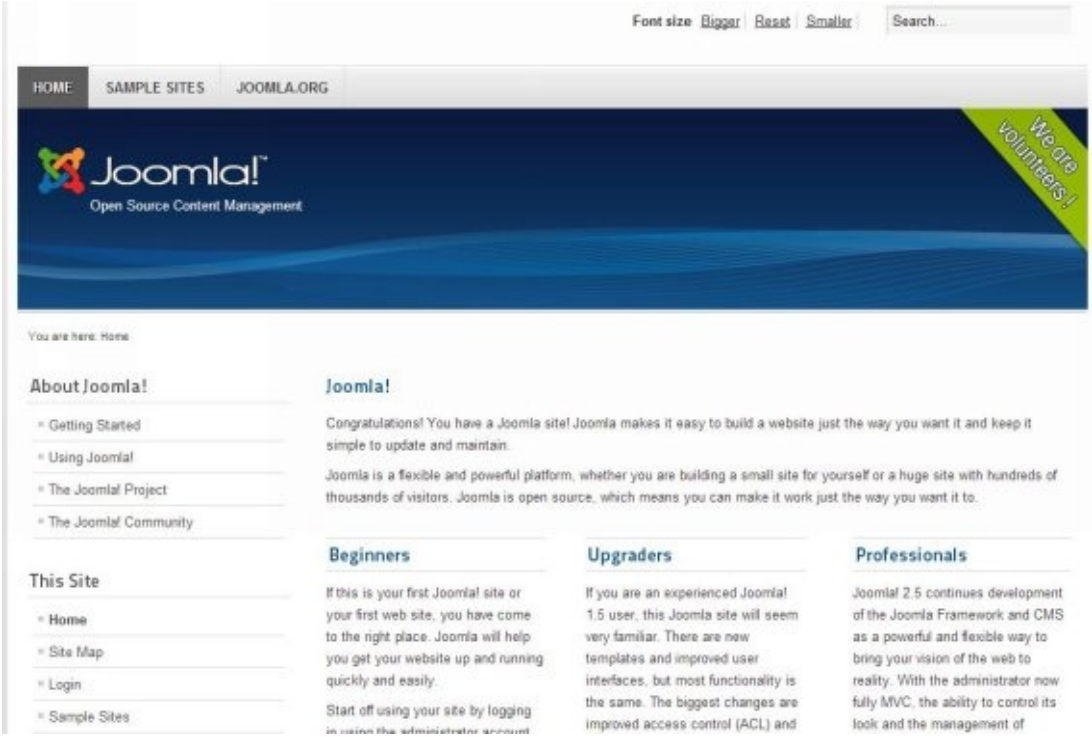
Whenever a Joomla site is opened, the following happens –

1. The website is displayed according to its code.
2. The look and feel comes from the Template, which is a bundle of organized CSS files.
3. The content of the web pages like texts, links, etc. are retrieved from the MySQL database.

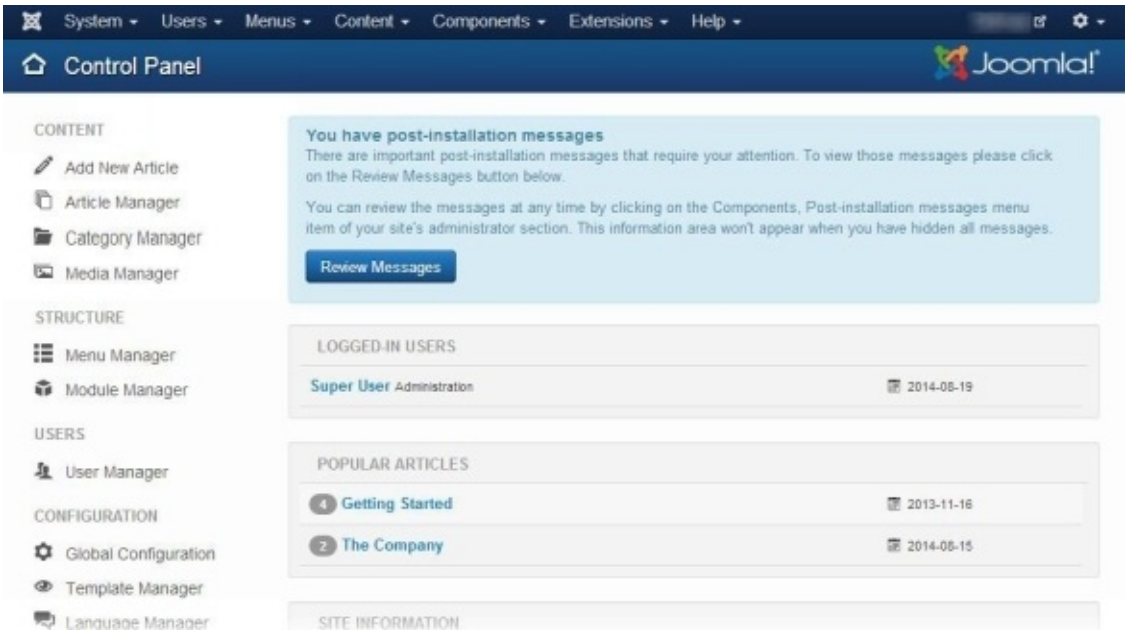


2.2.1 Joomla Website Screenshots

This is how a **Joomla 2.5** Frontend looks like right after installation –



Following is the Backend of a **Joomla 3.3.3** –



I have used a newer version to show the backend because, there's more to learn about the Backend. You can always alter the way the Frontend looks with use of templates or your customized styling. We will discuss about it later.

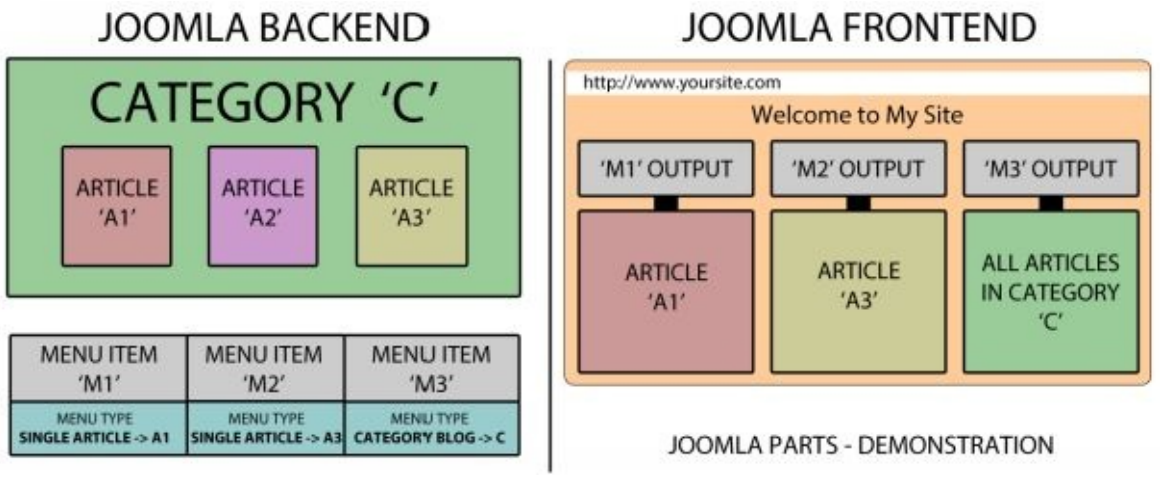
2.3 Parts of Joomla

To a beginner, the Joomla backend might seem to be like a huge library with scattered books but if you know how things are organized, it might prove to be the best resource bundle you could ever have for designing what you want.

Let us now understand the important parts of Joomla –

1. Category Manager
2. Article Manager
3. Media Manager
4. User Manager
5. Menus
6. Extensions

The figure below explains how Joomla parts are fitted properly to give a desired output –



As we can see in the figure, **Articles** like A1, A2 & A3 are created under a **Category** named C.

The 1st **Menu** item M1 is declared as a 'Single Article' menu type and is assigned to **Article** A1.

The 2nd **Menu** item M2 is also declared as a 'Single Article' menu type and is assigned to **Article** A3.

The 3rd **Menu** item M3 is declared as a 'Category Blog' which retrieves all articles under a particular category, in this case **Category** C.

So in the **Joomla Frontend**, when a visitor clicks on the respective menu items, the assigned menu item values are retrieved and displayed as an output.

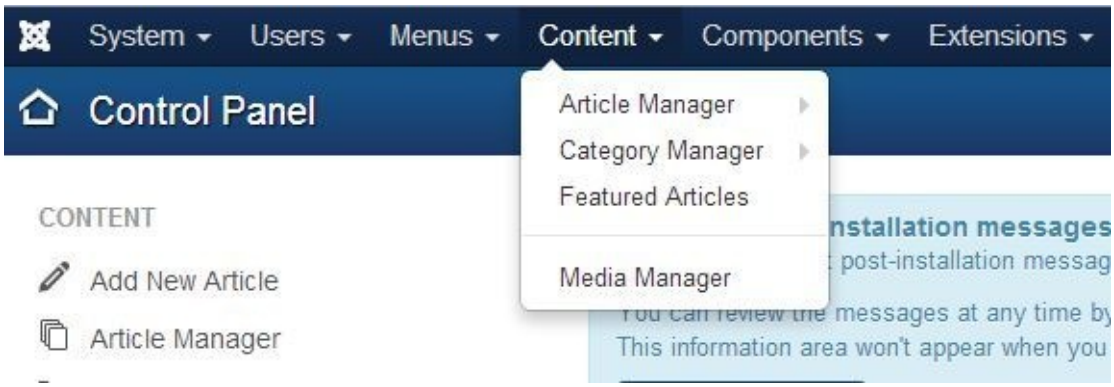
Now, the question is, how is the output displayed? Well, it is not displayed like a code but exactly the way an article has been originally written by the designer. This is possible by use of a Template extension. An article might also include pictures and other multimedia contents from any external URL or from local files as available in the Joomla Media

Manager of your site.

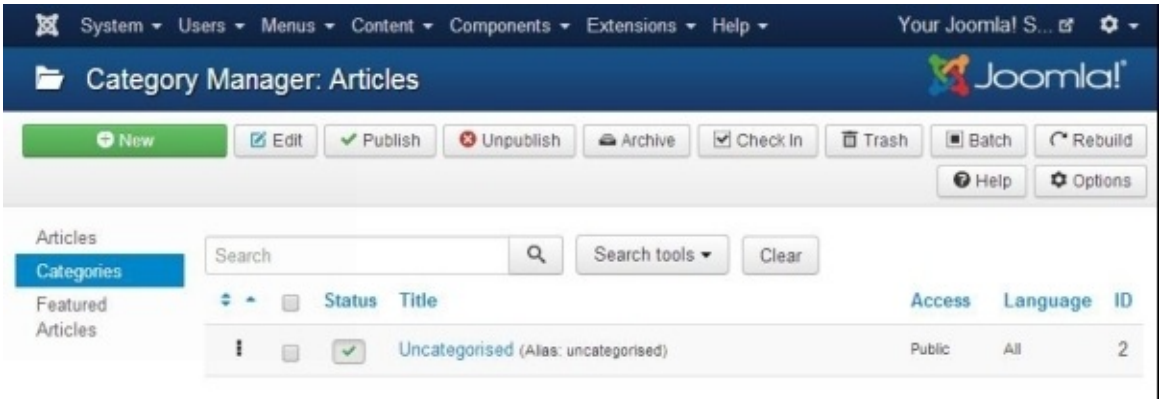
Now we will discuss these parts of Joomla in details.

2.3.1 Category Manager

A Category Manager can be found under the **Content** option in backend menu as shown below –



Here you can create multiple Categories or browse and edit existing categories. The Category Manager looks like this –



You can create a new Category by Clicking on the green colored **New** button on the left and edit preferences to either **Publish or Unpublish** respective categories. For example, here we can see a Category named 'Uncategorized' with a green tick appearing on its left. That means the category is published and any article written under this category will also be published.

In Joomla, the **Check In** option is useful when more than one administrators are working in the backend. This allows blocking an element from being edited by an administrator while it is being edited by another administrator.

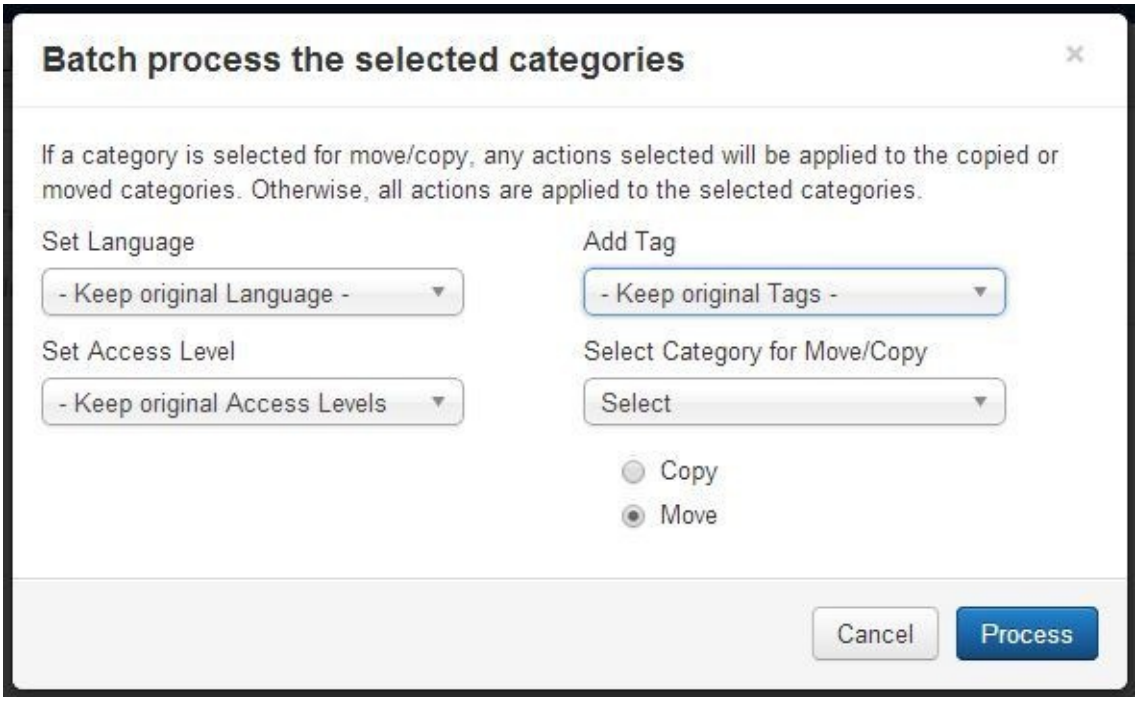
While creating or editing a category, you can restrict its access for certain user groups. For example, you want to let only the registered/ logged in users to view contents of a category. In that case, you can specify the **Access Rights** to Registered. Other categories with Public access like Uncategorized as shown in the figure above, can be viewed by non-registered or non-logged in users.

We will discuss about User Groups and Access Levels later in Section 2.3.4 of this book.

The **Archive** option is useful if you publish several articles and there are old articles that

you still want to be published on the website, but have been displaced in priority by new ones. Archiving helps in maintaining large databases of articles sorted by date, as done in blogs.

The **Batch** option is used to perform similar tasks with multiple objects, for e.g., if you wish to change access levels of all categories at once, you can update with a batch process as shown below –



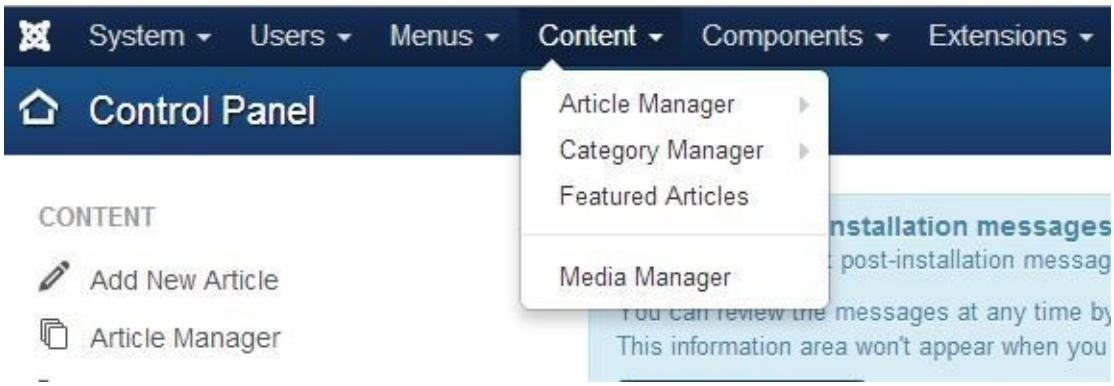
The Batch process saves time and effort for performing repetitive tasks. It is important to keep in mind that if you want to process selected multiple categories and not all in a list, then you must use the checkbox to the left of the categories to make a selection before Batch processing.

As mentioned earlier, like any other information, the categories and articles are stored in the MySQL database and Joomla backend provides us with an organized way to view and modify the database tables. So every Category appears with an ID to the right corner of the tabular display.

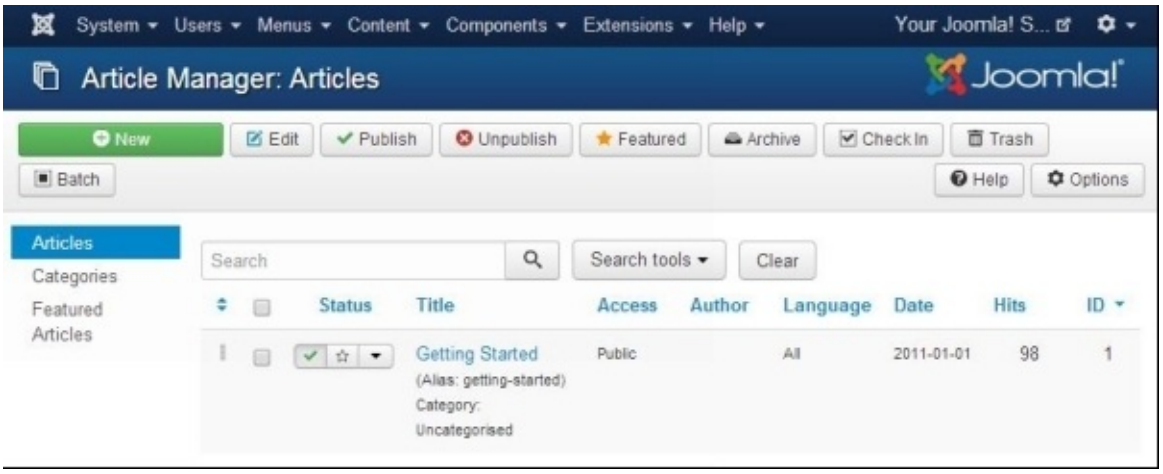
The ID acts like an identifier of a particular item in the respective database table. This system of unique identifying enables the relational database to connect multiple tables with one another for e.g., coupling a particular article with a respective category.

2.3.2 Article Manager

The link to the Article Manager appears right above the Category Manager as shown below –



An article manager lets you create and manage articles under different categories. An article manager in Joomla 3.x looks like this –



A new article can be created by Clicking on the New button on the left. Other functions like Publish, Unpublish, Archive, Check In, Trash and Batch work similar to that of Category Manager.

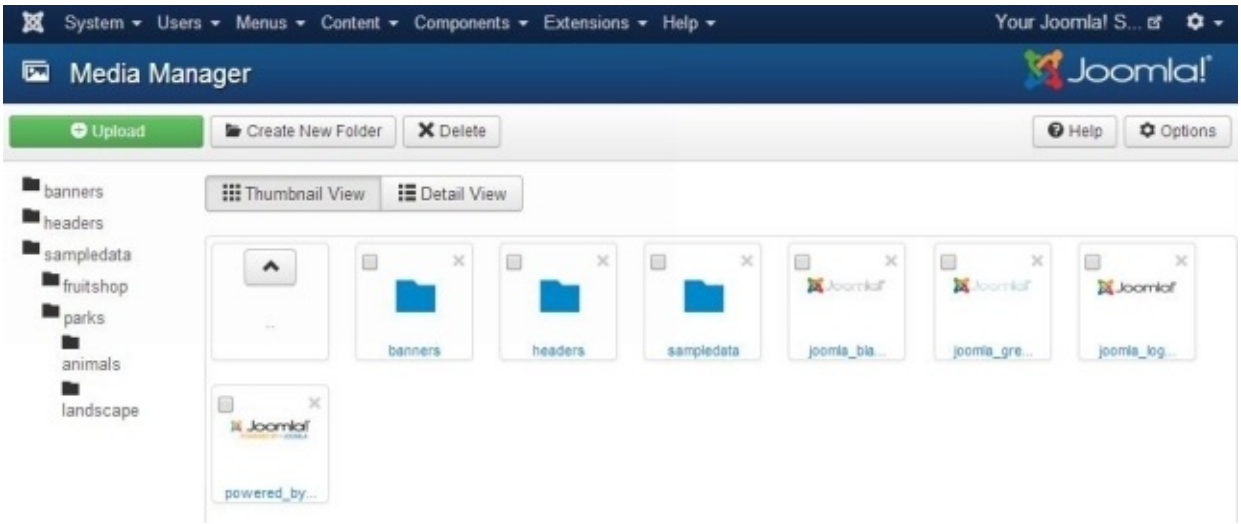
Joomla enables you to Feature selective articles and display them on pages you want. You can simply select an article by the respective checkbox and click on the Featured button above or you can simply click on the Star button appearing to the left of the article name in the article list, to feature an article.

Assigning featured articles to menu items will be explained later in this book when we start creating a Joomla website, step by step.

2.3.3 Media Manager

The Joomla media manager lets you upload and manage image and multimedia files to your Joomla website. It acts like a library where you can create sub directories according to your necessity and store files. While creating an article, you can browse the Media Manager to insert photos into your article.

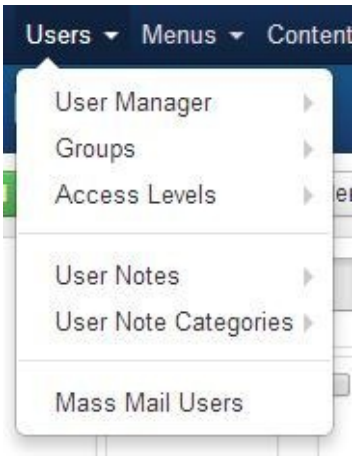
The **Media Manager** looks like this –



You can find the link to the Media Manager under the **Content** menu of your Joomla website, just like the **Article** or **Category Manager**.

2.3.4 User Manager

The **User Manager** in Joomla helps you to manage users registered on your website.



You find this option under the **Users** menu in the backend. The user manager helps you view registration details of users, their last login data, access levels and IDs.

You can also create new users with email IDs and activate their accounts so that they are able to login.

As you can see in the figure on the left, there are options like **Groups** and **Access Levels** just below User Manager. These options are really important for securing your website content from any unauthorized usage.

While Groups help you to classify your users for e.g., Administrator, Registered, Public, Editor, etc., the Access Levels option helps you to assign proper access rights to particular group of users. For instance, you wont let a visitor on your website to edit the front page text. But you might let an Editor to do it for you. So before letting them in, you have to specify who should have what rights on your site.

The **User Group** page is shown below –

Group Title	Users in group	ID
Public		1
|— Guest		9
|— Manager		6
|—|— Administrator		7
|— Registered		2
|—|— Author		3
|—|—|— Editor		4
|—|—|—|— Publisher		5
|— Super Users	1	8

From the above image, you can understand how user groups are classified according to their level of interference in the website.

Let us now understand how access levels are defined in Joomla. Here is a screenshot of the Access Levels page –

Search

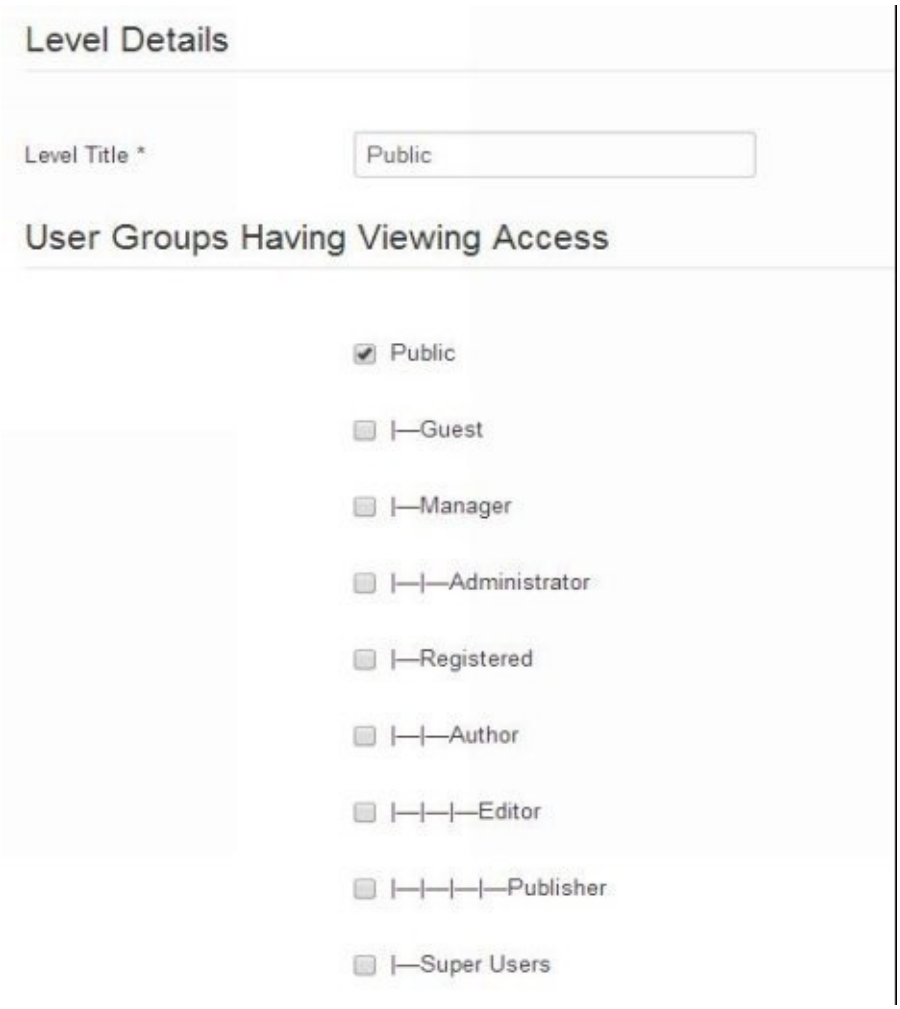
Level Na...

Level Name	
Public	1
Registered	2
Special	3

This is the page that opens up when you click on the **Access Level** option under the **Users** menu. Here we can see 3 different access levels have been defined – Public, Registered and Special.

A **Public** user is a normal non-logged-in visitor. When a public user logs in, the group changes to **Registered** or **Special** depending upon scenarios that you would define for your website. For e.g., you might have different access rights for teachers, students and coordinators. In that case, you can well utilize the access levels to restrict or allow operations by a particular user groups when they are using your website.

Most of us often confuse Groups with Access Levels owing to similar naming pattern. To overcome this, let us click on ‘Public’ to learn how the correct access rights are assigned –

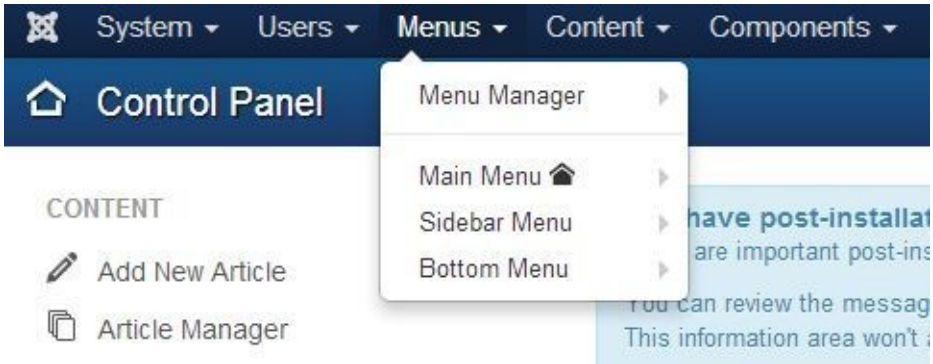


Here we can see how a Public Group is assigned to have access to only the items which are declared as Public in the Joomla backend. For e.g., the front page article is Public. So a visitor is able to view it, however, the Public visitor is unable to view any page in the private or members area of the website until he logs in and his group changes from Public to Registered.

You will find the use of Access Levels, an important thing while making members area on your site and restricting certain categories, articles, menus and modules from getting viewed by non-members. Such options are available under the ‘Permissions’ tab.

2.3.5 Menus

By now you must have certainly understood that most of the common tools under various parts of Joomla work in a similar manner. So let us now learn about one of the most important parts of a website – the menus, which enable us to create a navigation system for our website. In Joomla, the **Menu Manager** can be found under the **Menus** option as shown in the figure below –



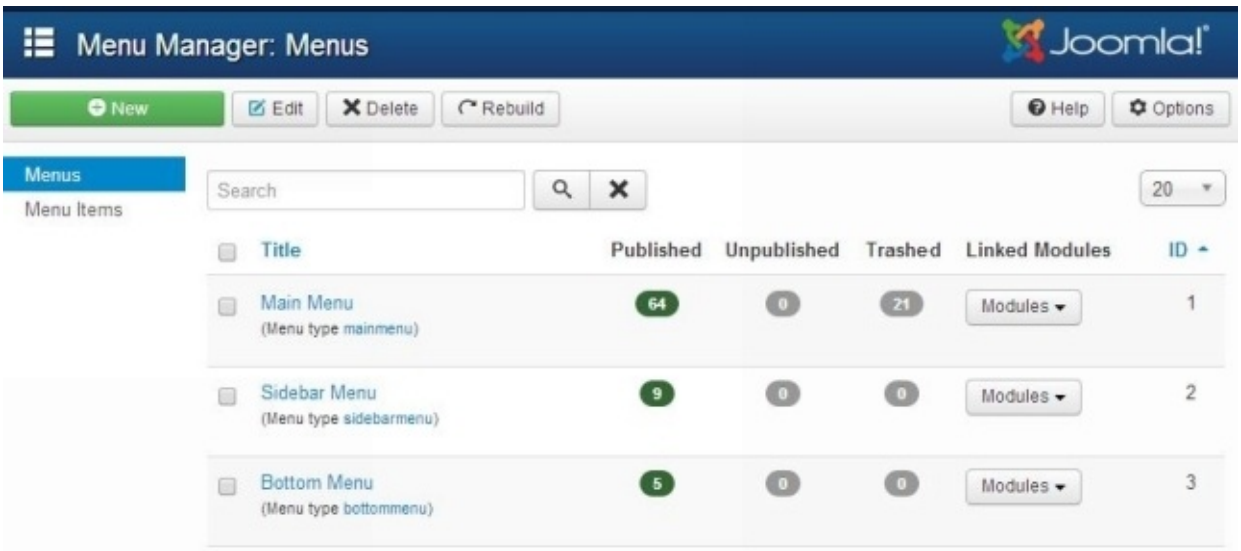
Now from here we have choices to make – 1. Edit or Add Menus; 2. Edit or Add Menu Items under a particular Menu. But before making a choice, first let us understand what the difference between the two options is.

Menu – it is a collection of Menu Items for e.g., Main Menu may contain items like Home, About Us, etc.

Menu Item – it is the hyperlinked item under a parent menu which, on an event of a click or tap leads the user to a defined link / page or performs a desired function. As already stated above, example of a Menu Item can be Home, About Us, Services, etc.

You might be thinking, why to have more than one menu when all menu items can be put under a single menu. However, the scenario is not always the same. For example, you want the Home, About Us and Services to appear on the top of the website and you wish to place menu items like Advertise, Careers, etc. at the bottom. In that case it is a good practice to create a Main Menu for the top and a Footer Menu for the bottom of the website. Likewise, while creating members area, you can place different groups of menu items at separate positions on a page for logged in users making the navigation clean. Such clarity of navigation requires assigning of multiple menus for better visibility and ease of access.

So coming back to the choices, let us first open the Menu Manager to check out the already available menus and then we shall proceed into further details. Therefore we now click on Menu Manager and the following page opens up –



Here we see 3 menus – Main Menu, Sidebar Menu and Bottom Menu.

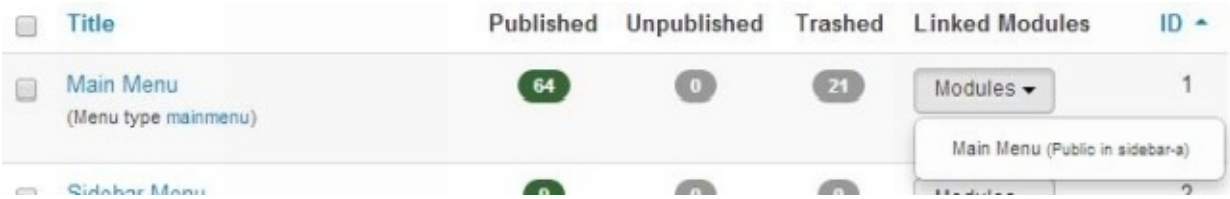
The **Main Menu**, as the name suggests, consists of the main links or menu items and is generally placed near the top of a webpage to serve as the main navigation of a site.

The **Sidebar Menu** which is an optional menu, might have members - area items and as the name suggests, it must have been placed somewhere by the sides of certain web pages.

The **Bottom Menu**, as we have discussed earlier, appears to be a footer menu consisting of footer links.

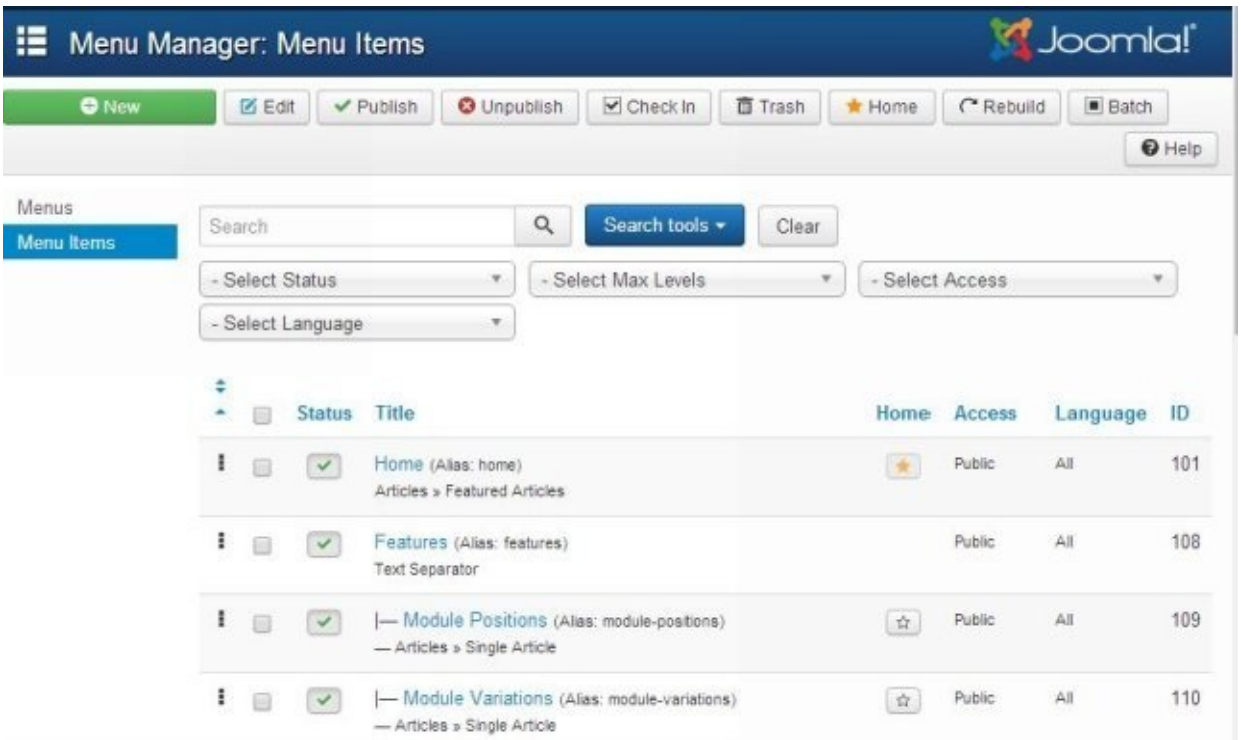
Now if you notice, each Menu has got a number Published items, as shown in the figure above. The number appearing in the green rounded container is the exact number of menu items corresponding to each menu. Likewise, Unpublished and Trashed menu items are also displayed.

But the question is, how are the menus displayed on the Frontend? For this we require Modules, about which we shall learn in the next portion of this chapter. But for an overview, let us see the image below –



Here, the Main Menu is linked to a Module which has Public access rights and has been placed in ‘sidebar-a’ position of the website template. Although there is another way of assigning a menu to a module, it is always good to know about possibilities.

Now we will click on Main Menu to check out the detailed page that is contained within.



In the image above, we can see Menu Items like ‘Home’, ‘Features’ and sub-menu items like ‘Module Positions’ and ‘Module Variations’. Notice the Yellow star that appears to the right of ‘Home’. This means the Menu Item ‘Home’ is the default page or the Home page of the website. The type of the menu is set to ‘Featured Articles’ as shown in the image just below the ‘Home’ menu item. So as explained earlier, you can understand that this menu item is set to display only the Featured Articles from a particular category of articles.

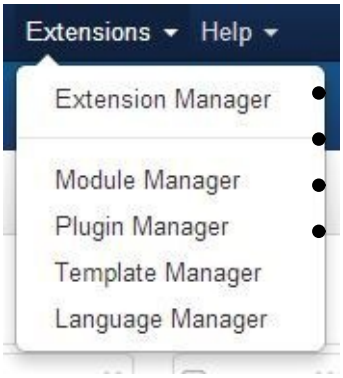
The **Access** level is set to Public – obviously you won’t wish to annoy your visitors by blocking the access to the home page, asking them to log in without even knowing what your website is all about.

The rest of the tools that appear on the upper side of the page like search filters and buttons are almost similar to that of what we have learned in Category or Article Managers. You can try playing around a bit while you are testing your Joomla website.

2.3.6 Extensions

In Joomla, the default website is useful for creating a basic website smoothly. However, if you wish to add more functions and features to your existing Joomla website, you can install **Extensions**.

Extensions are packages of codes written to perform certain added functions which the basic Joomla website cannot. For example, you wish to turn your website into a Shopping portal. In that case you can install certain ecommerce extensions, which can add new features to your existing site like customer accounts, shopping carts, payment gateways, product specification pages, etc.



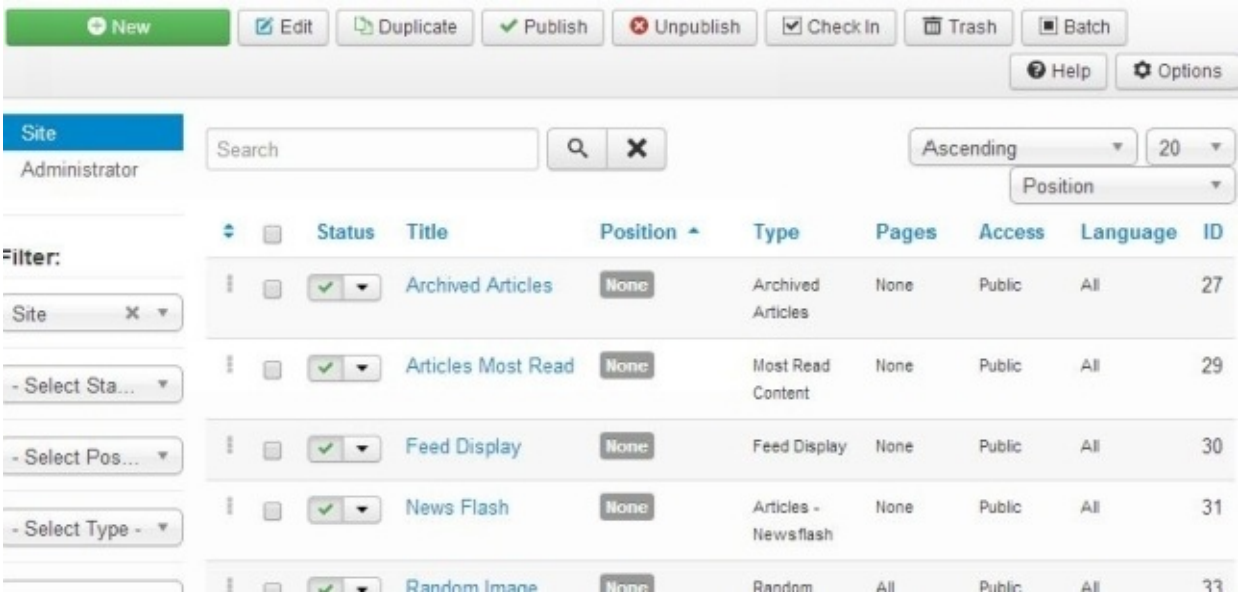
Extensions can be of four types –

- Modules
- Plugins
- Templates
- Languages

However, **Components** can also be considered as Extensions, but the Joomla backend identifies the above listed items as Extensions, and it has created a separate backend menu item for Components, owing to the complexity and scope of such extensions.

It is necessary to understand the meaning of each type of extension.

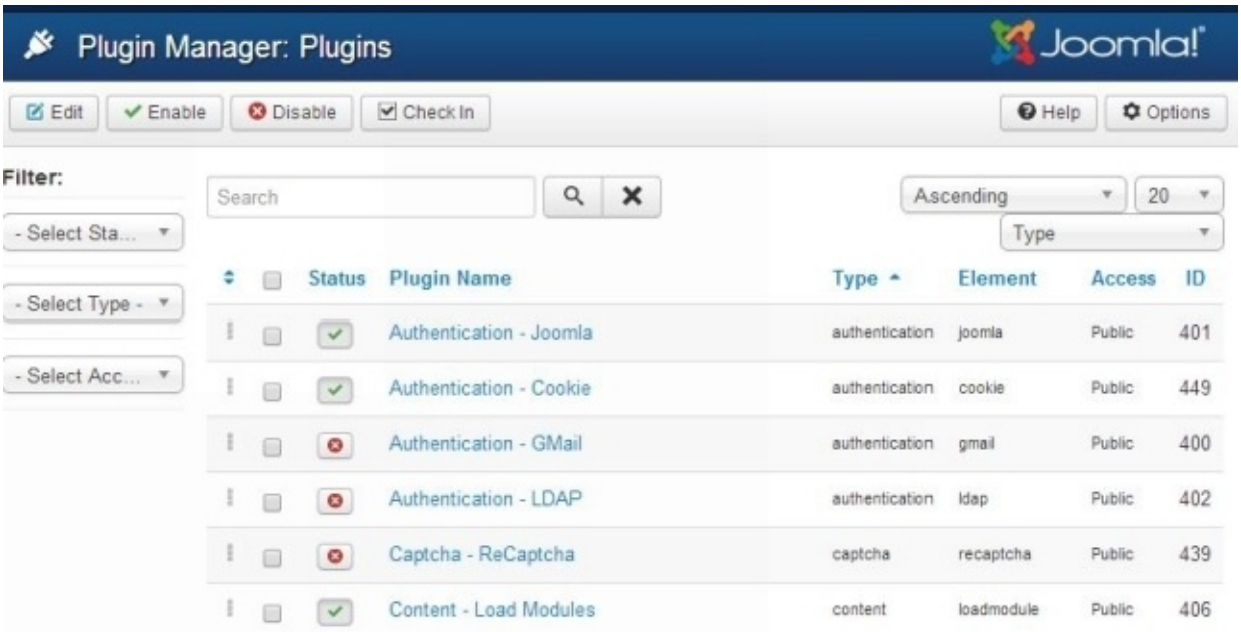
Modules – sections or panels appearing on a website's particular position, performing certain tasks. For example, you can use a Calendar module to display an interactive calendar on a particular area of your website. Modules are useful for placing specific utilities on a webpage without hampering the basic function of the page.



Modules can also be used for extensive styling, for example, using a module to display a series of videos with automated sliders near the bottom of your webpage. If used smartly,

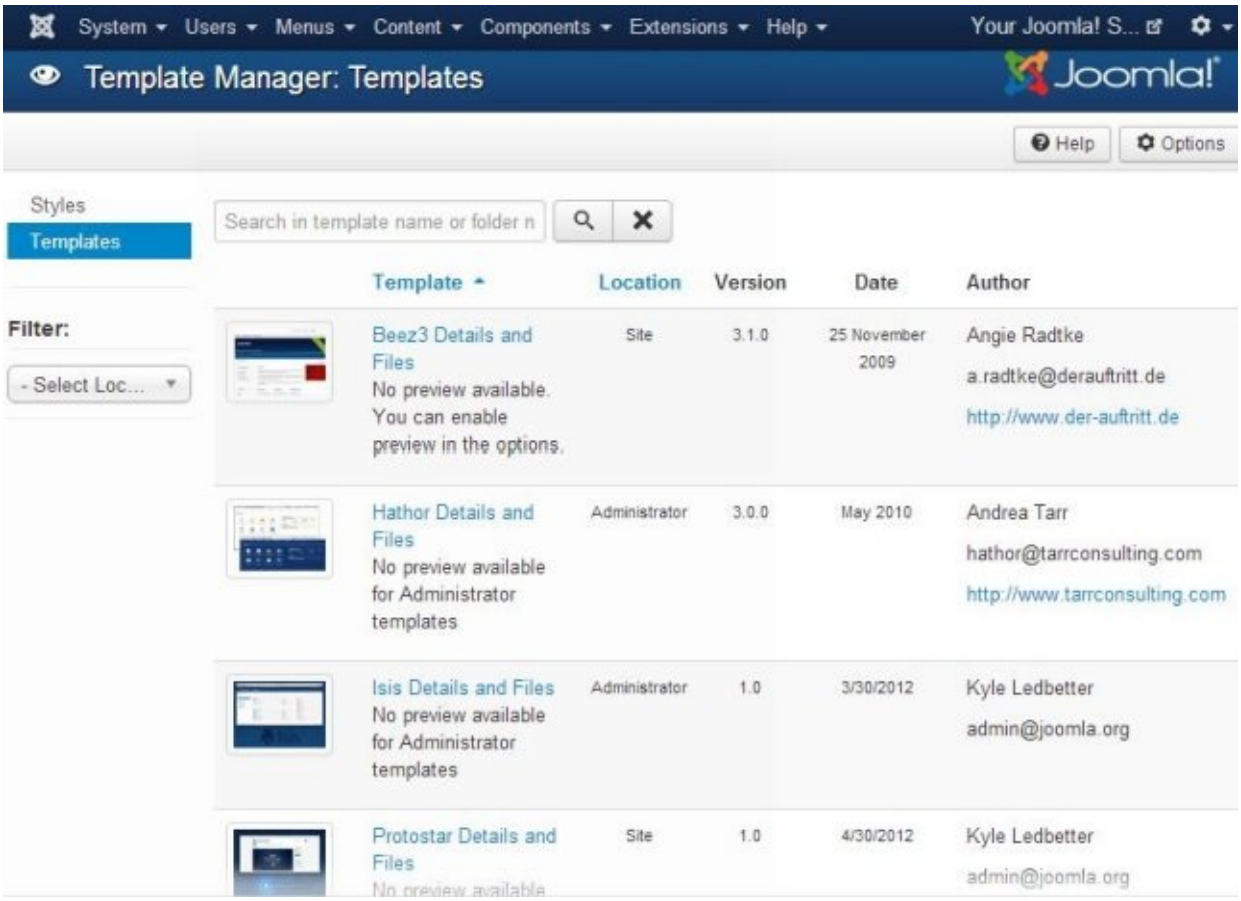
modules can really add not only to the function but also to the overall presentation of a webpage.

Plugins – as the name suggests, these are small extensions of the basic Joomla website operations. Plugins are like injections to a current set of code. One of the most common plugins already available in a standard installation of Joomla is the Read More button that appears below the default text editor while you create an article. This Read More button, when activated at a particular line, creates a Read More link to the current article and limits displaying the introduction of the article on any other page of the website. The user cannot view the full article until the Read More option is clicked. So this is a very small yet useful function. The Image, Page Break, Article buttons that appear along with the Read More button are also similar plugins. You can add more plugins like Gallery or Youtube to insert multimedia and interactive content into your articles. Apart from use in articles, plugins can be used to perform more critical tasks like security of the administrative interface, adding tools to an existing text editor, registration options management, library tools, etc.

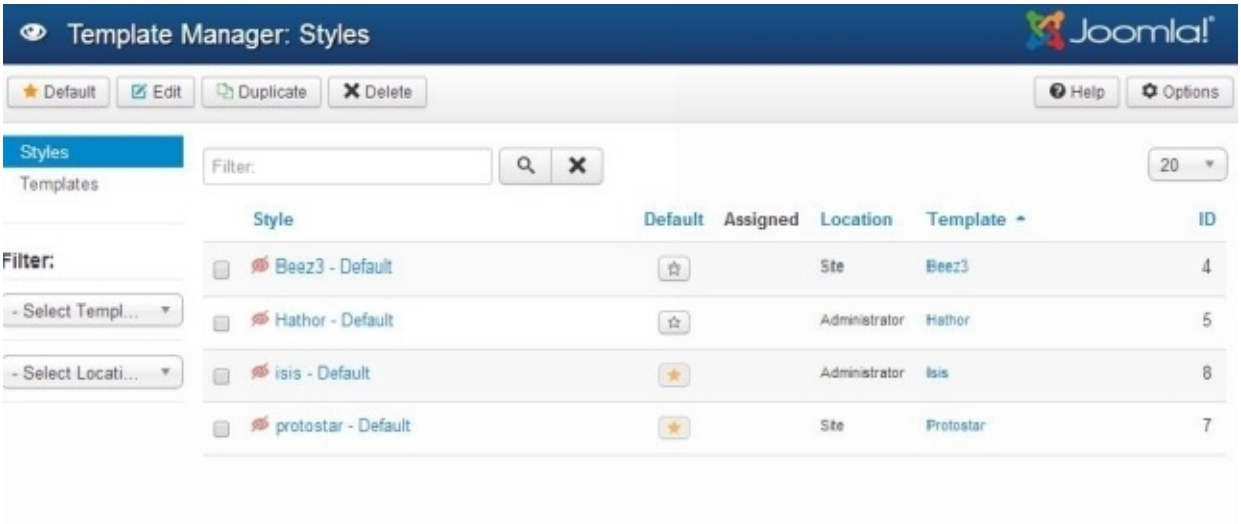


A plugins can affect the core System of Joomla or a particular Component depending upon the type of plugin it is. You can find this out when you open a plugin manager. If a plugin affects the core system, a text mentioning ‘System’ appears next to the name of the plugin, else the name of the component is written.

Templates – pre-defined styling for your website. A template can be of two types – Site Template & Administrator Template. We generally do not mess with the Administrator Template as we really do not often need to change how the Joomla Backend looks like. Most modifications or new styling is applied to the Frontend or the Site Template. A template can have multiple styles, for example, the homepage of your site displays a red border, while the other inside pages display a blue border on the same template. In that case, you can simply assign the **menu items** to the particular styling you wish to apply to the respective pages.

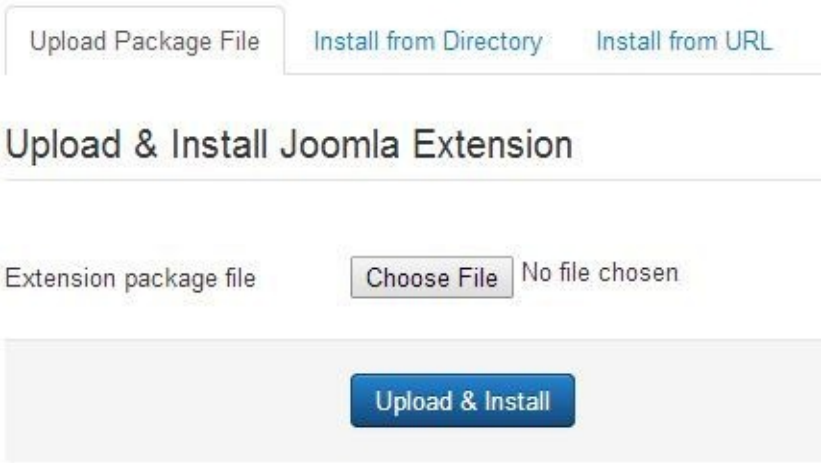


This a view of the template manager which shows four templates already present in the Joomla installation. Out of these, as we can see, two are Site templates and the other two Administrator templates. Now we will look at the list of styling presets these templates have. You can navigate from the Left Menu which shows two options – Styles & Templates.



From this image, we can understand that the Administrator default template is set to “isis – Default” and the Site default template is set to “protostar – Default”. This is similar to setting the homepage of the website by Clicking on the Yellow star in the Menu Manager. If you create more styling on the existing templates by editing the current styles, you can easily save the new styles with new names and the same shall appear on this Style list.

You may upload new Templates or “themes” as they are popularly known, through the Extension Manager.



Once successfully uploaded, the default styles & template will be appearing on the Style & Templates pages of the Template Manager. Then you can set the styling as the Default styling of your website by clicking on the ‘Yellow Star’ button under the Default column of the Style page under the Template Manager.

Few templates require a pre-installed Framework and generally the free download link to such framework is provided in the Template’s manual or the post-installation page. If you stumble upon such a thing, don’t worry. You just need to download the Framework package and upload it through the Extension Manager.

3.0 CREATING A WEBSITE

Creating a website with Joomla! involves installation of the Joomla package on the web server followed by creating the content you want for your site. Yes, it is that simple. However, this whole thing is achieved with the help of the Parts of Joomla we have learnt in the previous chapter.

3.1 Installing Joomla!

Before getting into installing Joomla!, you need a web server to install it to. If you are using your computer, you can use a virtual web server like XAMPP to run the website offline. You can watch the video on how to install XAMPP on your Windows 7 computer from the following link: https://www.youtube.com/watch?v=bKA5YVud5ZY

Installation method on Windows 8 is also the same.

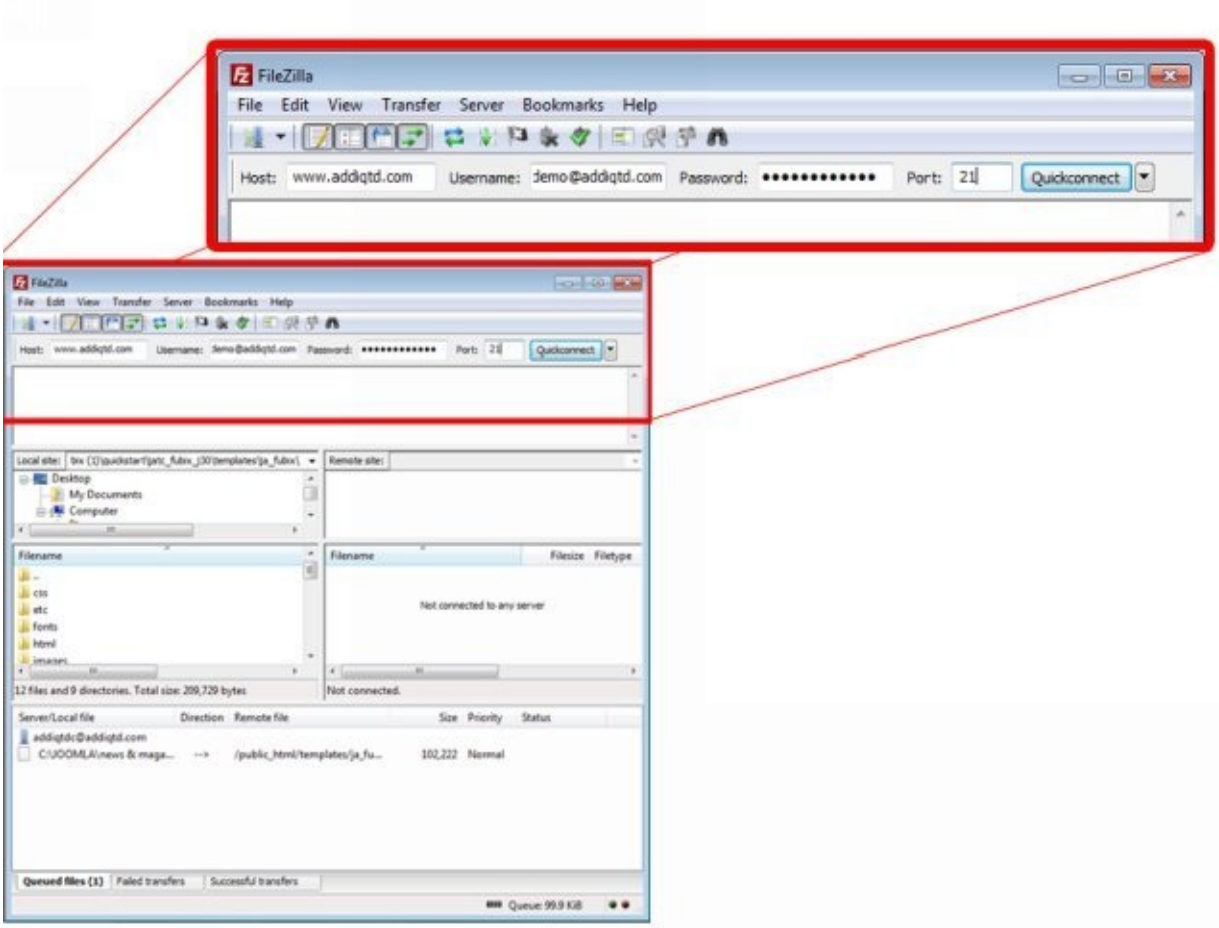
3.1.1 Creating a Database

Coming back to installing Joomla! on a web server either on your PC or the remote web server offered by your hosting service provider – first you need to **create a MySQL database** from the Database Wizard (if you are using a cPanel or any other similar hosting service) or create a Database directly from the MyPhpAdmin. **Create a new user** for the database and assign **all privileges** to the user so that once the connection between the Joomla website and the database is made, you can perform all kinds of read & write operations on the database through the Joomla site. Make sure that you set a strong **password** for your database which is hard to crack.

3.1.2 Setting the Website Path

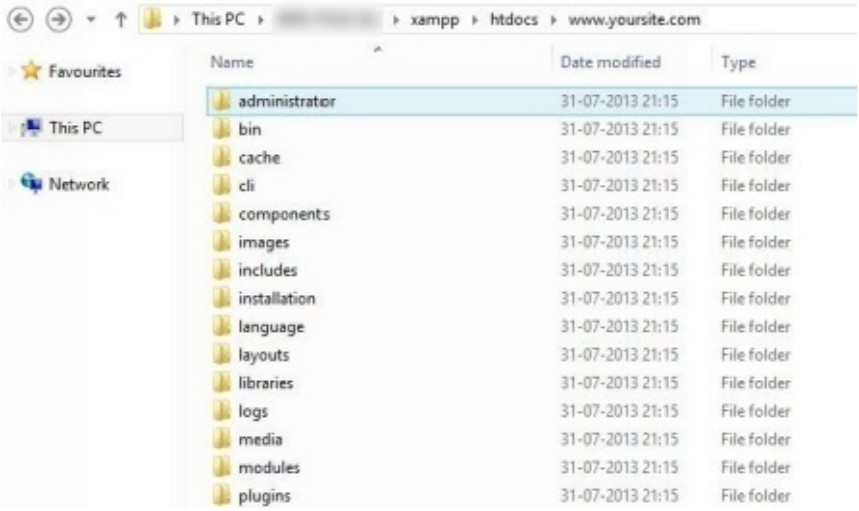
It is very important to understand the path where you want your website to be located. If you want your Joomla site to be accessible at *www.yoursite.com* then you have to extract the Joomla package into the root directory of your website. This means that opening the **root directory** of *www.yoursite.com* through a file manager or windows 'htdocs' folder (in case of XAMPP) should open up all the parent directories contained in a Joomla package like – Administrator, Images, System, Logs, etc. However if you want your Joomla website to be located in a **sub-directory** such as *www.yoursite.com/joomla* then the Joomla package has to be extracted in that particular sub directory, '/joomla' in this case.

You can use a FTP software like FileZilla to upload your Joomla package to the remote server for further installation of your site. For example, here we want to upload some files to our existing website www.addiqtd.com through FileZilla –



Here we have entered our **host name** as *www.addiqtd.com* and **username** as *demo@addiqtd.com*. These credentials should match your FTP configuration in the remote web server. Make sure that you have a username & the same password assigned to the FTP directory you are uploading your files to. Finally, you need to provide a **Port** number through which the connection shall take place, *'21'* in most cases as this one.

We have set up the installation on a localhost using XAMMP under www.yoursite.com:

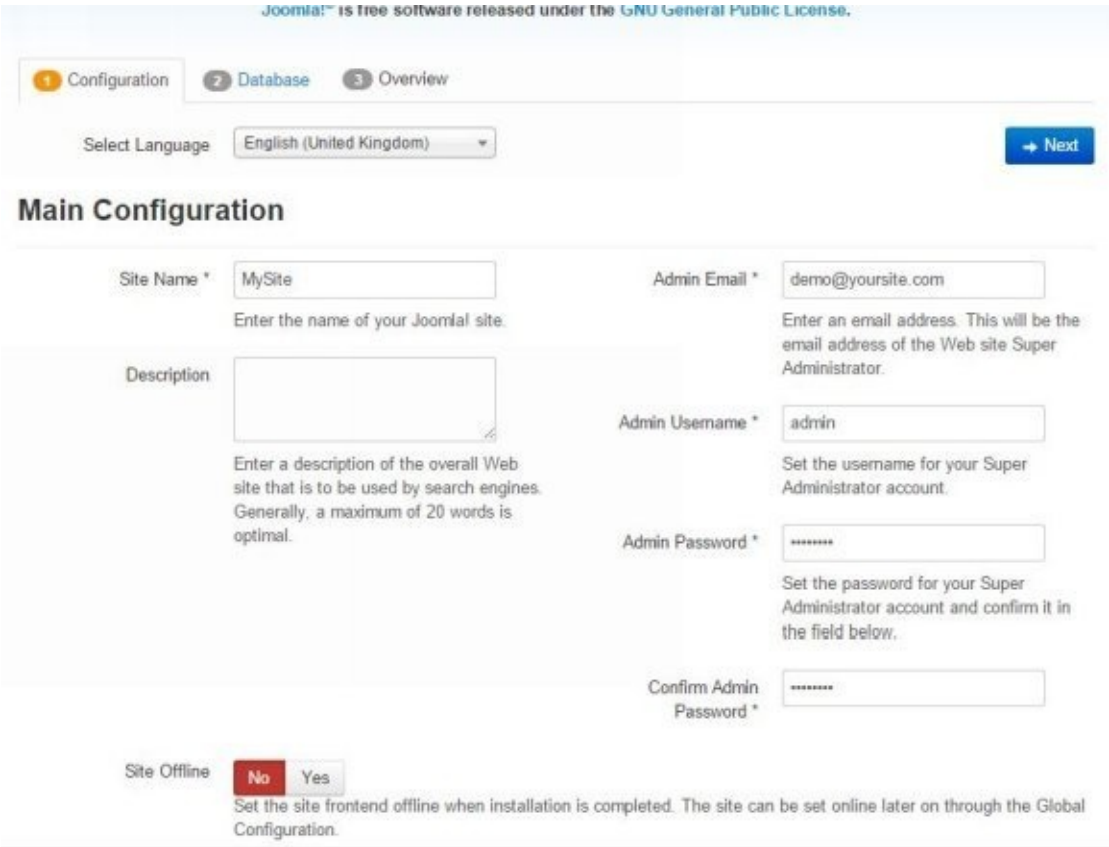


3.1.3 Configuration of Joomla Site

Now that you have created a database and uploaded the Joomla package into the desired path or directory, we will now run the installation of the Joomla site. To do that, we need to open a web browser like Google Chrome and type in the URL of our Joomla site, for

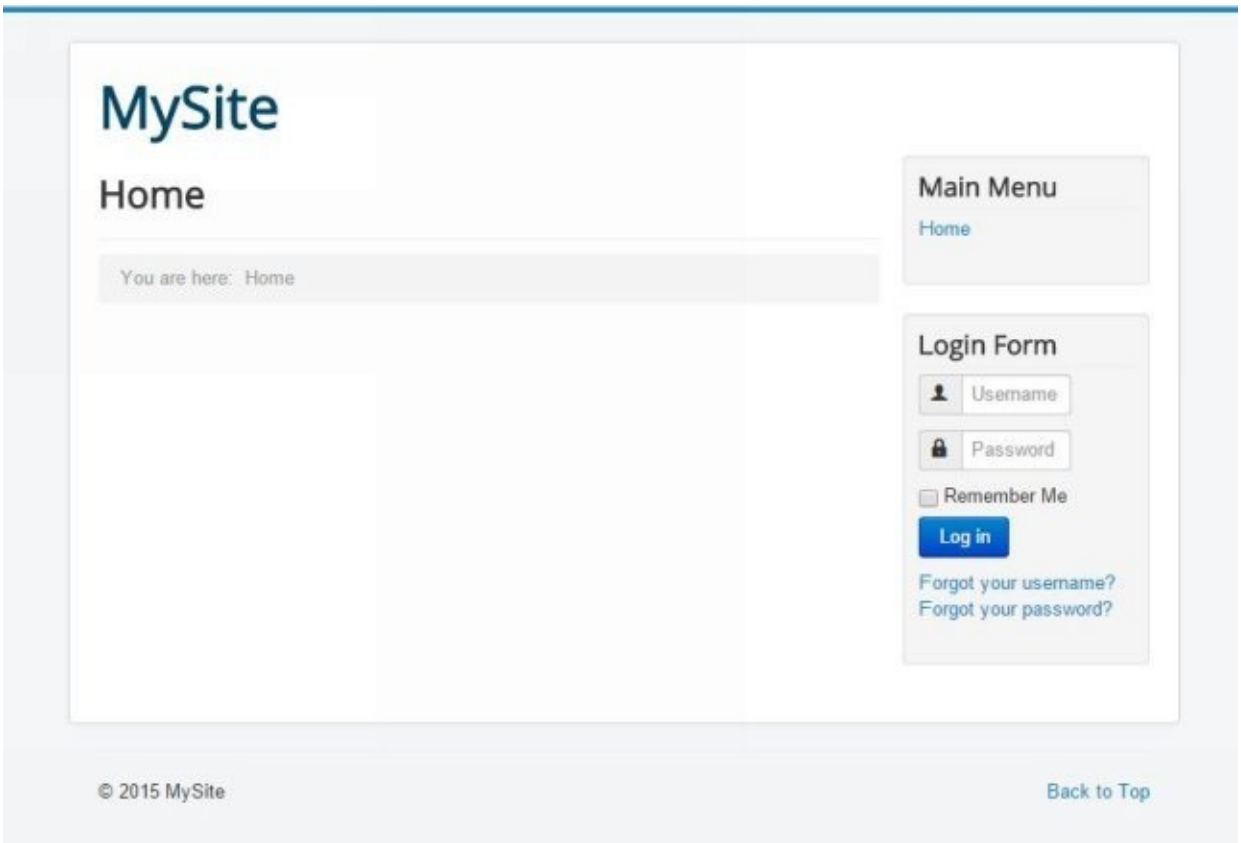
example, *www.yoursite.com* and this will automatically redirect to the installation folder of the Joomla package and run the web installer for Joomla.

Here you have to fill in the required details regarding your website like the Site Name, Database details, Administrator Username, Password, Default Email ID, etc.



Proceeding to the Next step enables us to fill up the details of the database which we have created in the first step & finally we click on the Install button to get our Joomla site ready.

The Joomla site takes just a few seconds to install and it is essential to remove the ‘Installation’ directory after the installation is done. Here’s how our **Joomla 3.4.2** site named as ‘MySite’ looks like –



Now it is time to add some content to this website and make a basic business website.

3.2 Navigation & Content

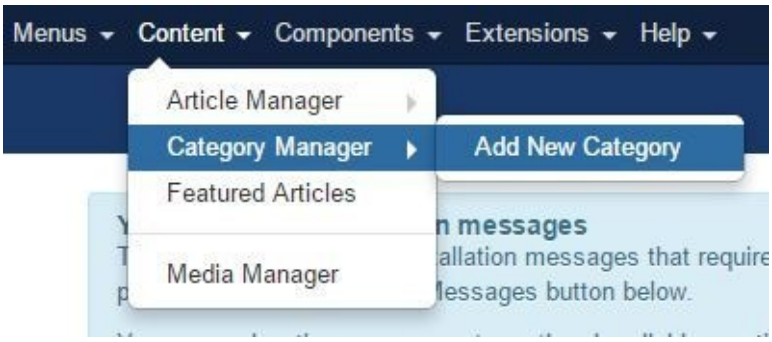
A basic planning or prototyping of the website on a paper or white board is useful before plunging into designing a website. Therefore, to begin designing a basic business website, we shall consider the following pages to be built on the site:

- Home
- About Us
- Services
- Contact Us

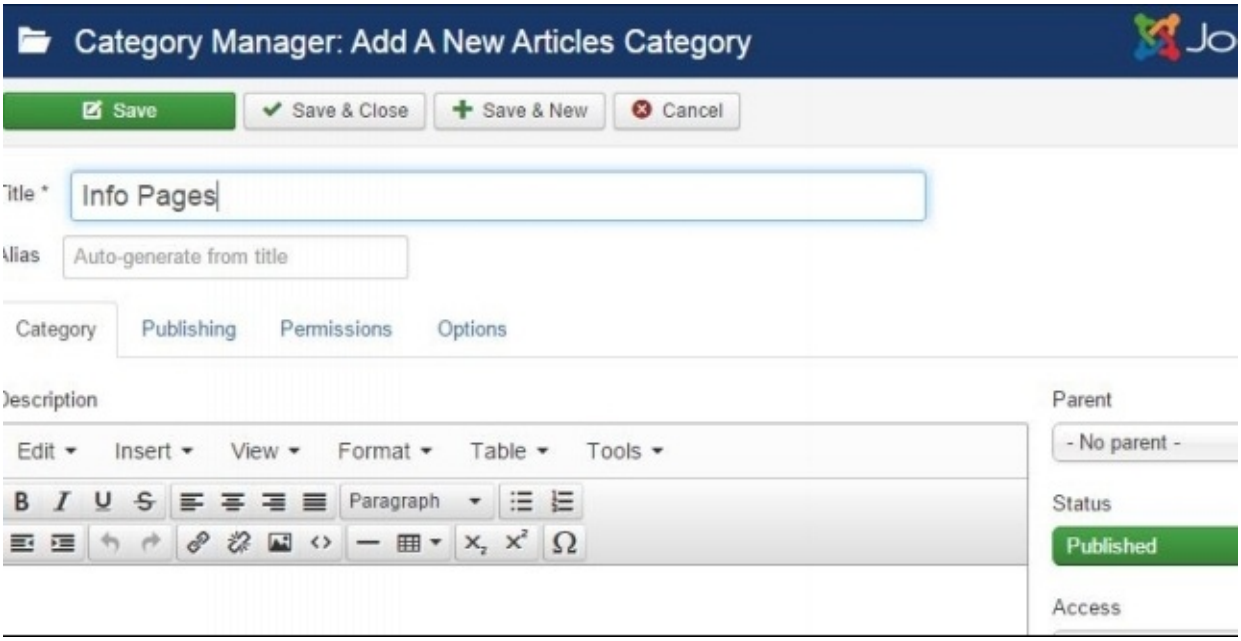
Please keep in mind that we are going to make this website for demonstrative purpose only, and you are free to try it yourself and improvise with the design according to your imagination and choices. There are endless possibilities.

3.2.1 Creating Categories

After logging in to the Joomla backend from *www.yoursite.com/administrator* we navigate to Content Category Manager Add New Category to add a new category under which we will be adding new articles.



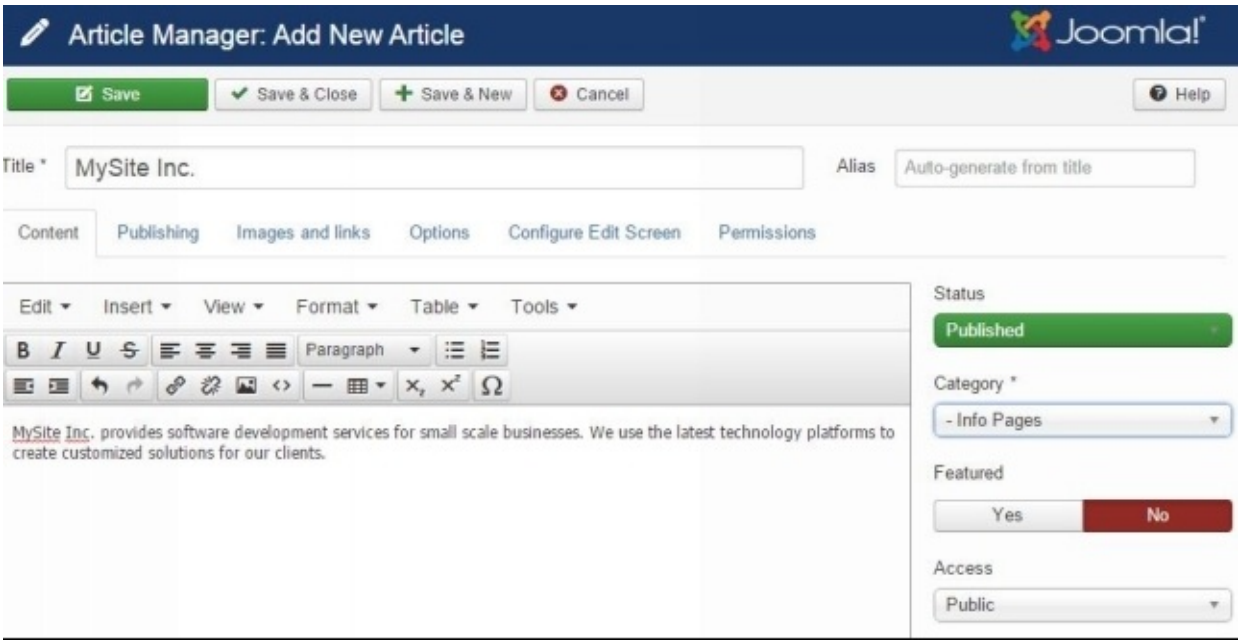
We add a new category 'Info Pages' to add informative articles related to the business and click on Save to store the category into the database. Make sure that the status is set to 'Published', else the articles under the category won't be visible on the website's frontend.



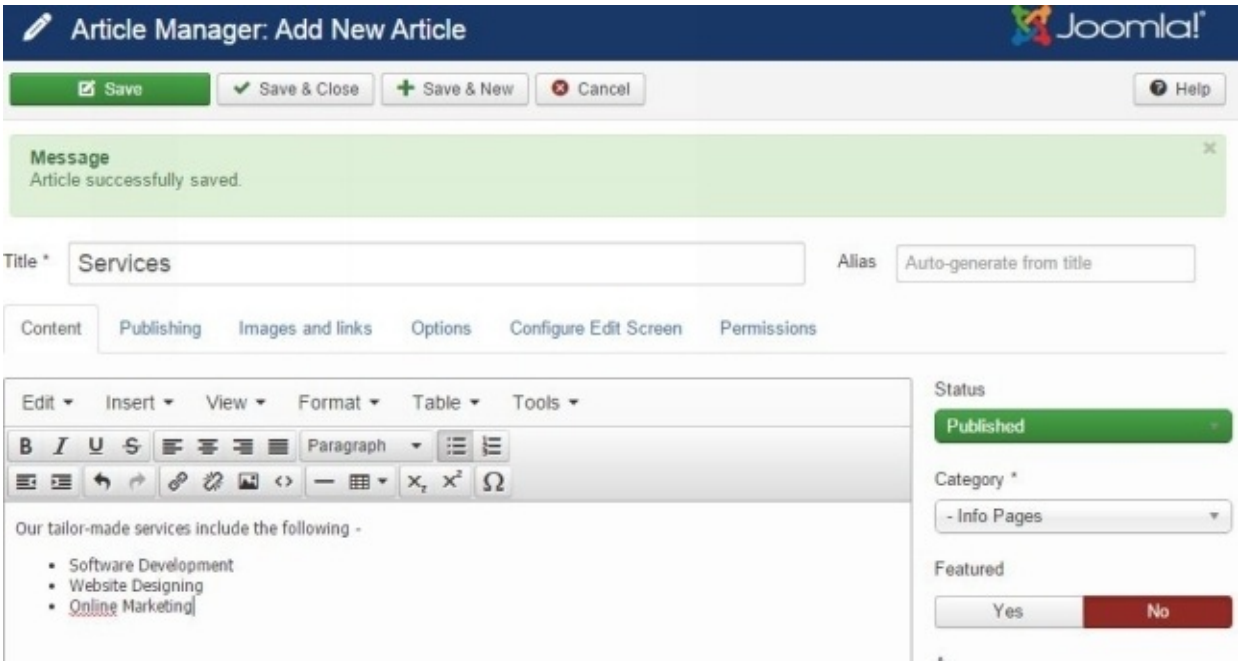
3.2.2 Creating Articles

You need to understand a concept in Joomla. The web pages we see on the site's frontend are actually articles in the backend. Then why don't we just call it a Page? This is because, an article stores the text, table and graphic data in HTML format. This actually gets stored in the database and doesn't get displayed on the site as a page until and unless it is assigned to a Menu item. The article can be displayed as a single page or it can be displayed as a post in a blog. The same article can also be used to display a slideshow of content using plugins and short codes. So, it is wise enough to call an article by its name.

Now we add some articles to match our initial plan for the site. We navigate to Content Article Manager Add New Article.



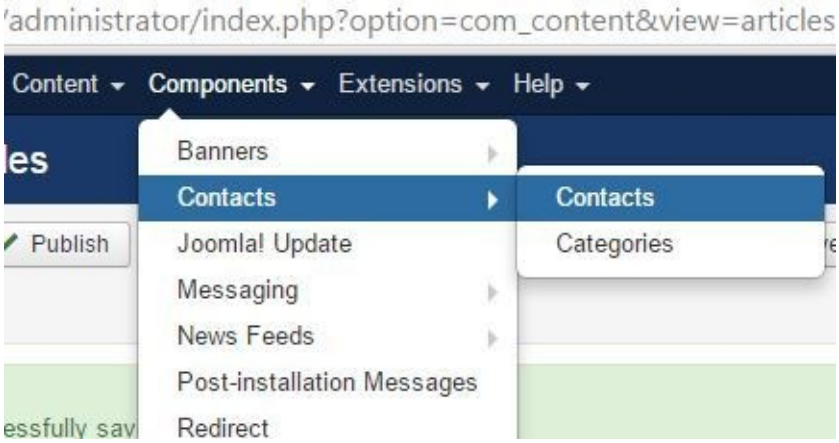
Here we add an article titled as ‘MySite Inc.’ and add some text content to the article under the Content tab in the text editor. We assigned the article to the Info Pages category from the right panel. Also make sure that the Access is set to Public, so that any visitor on the site is able to view the article without having to register and login. Finally we click on the Save & New button and proceed to adding another article named as ‘Services’ in the same manner. After writing the content, we click on Save & Close, as we don’t need to add any more article at this time.



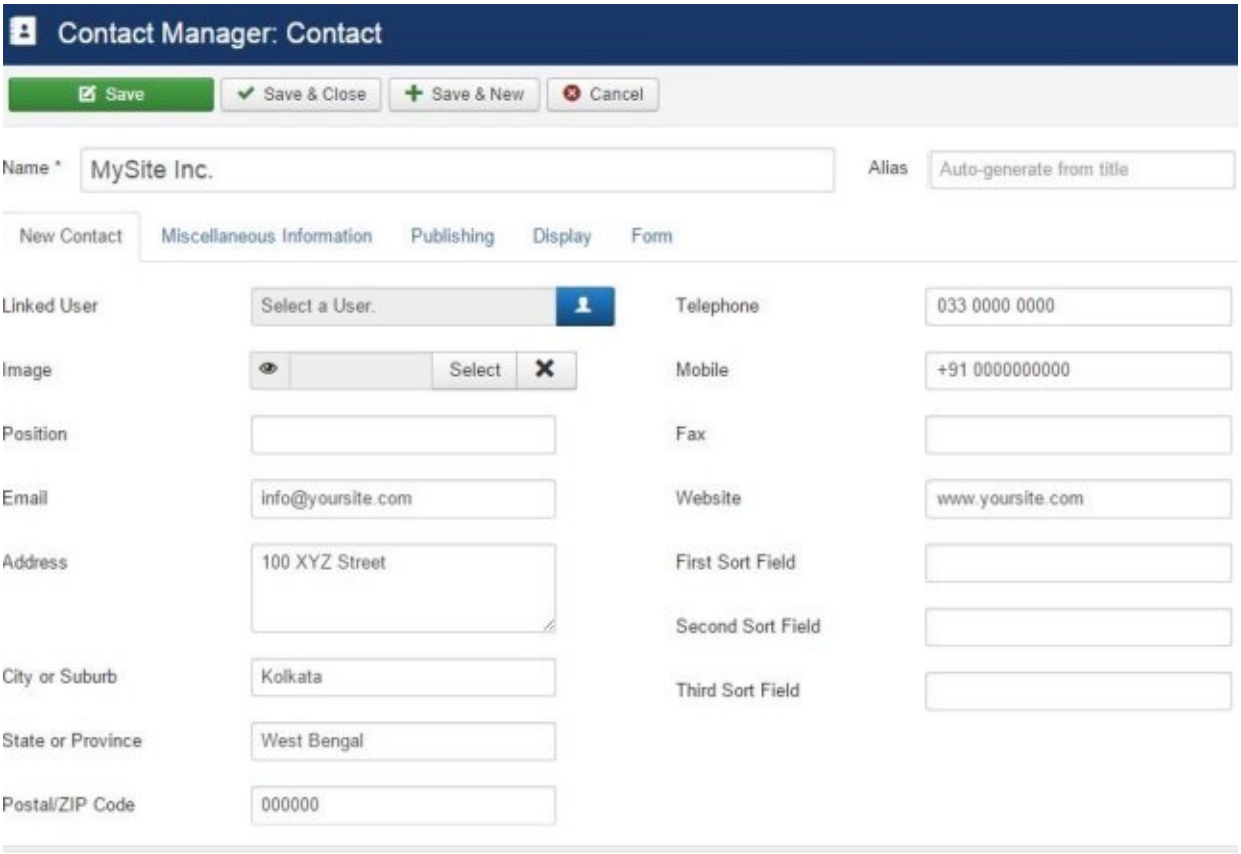
Now that we have prepared the content for two pages of the website, we need to prepare a contact page which can also display a contact form along with all contact information of the business.

3.2.3 Adding a Contact

To create a contact page, we first need to navigate to the Contact **component** andadd a new contact. To do this, we go to Components Contacts Contacts.



From the next page, we click on New and fill up the fields in the contacts component as shown below:

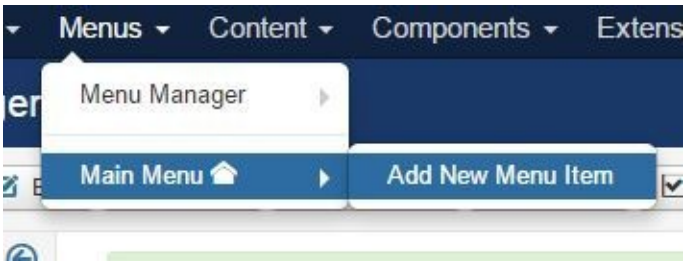


3.2.4 Creating the Navigation

Now we start creating the most important thing for the website to be accessible properly. Till now there has been no visible change on the website's frontend. Without navigation, the visitors won't be able to browse the website. Therefore we need to create a Menu with Menu Items to serve the purpose.

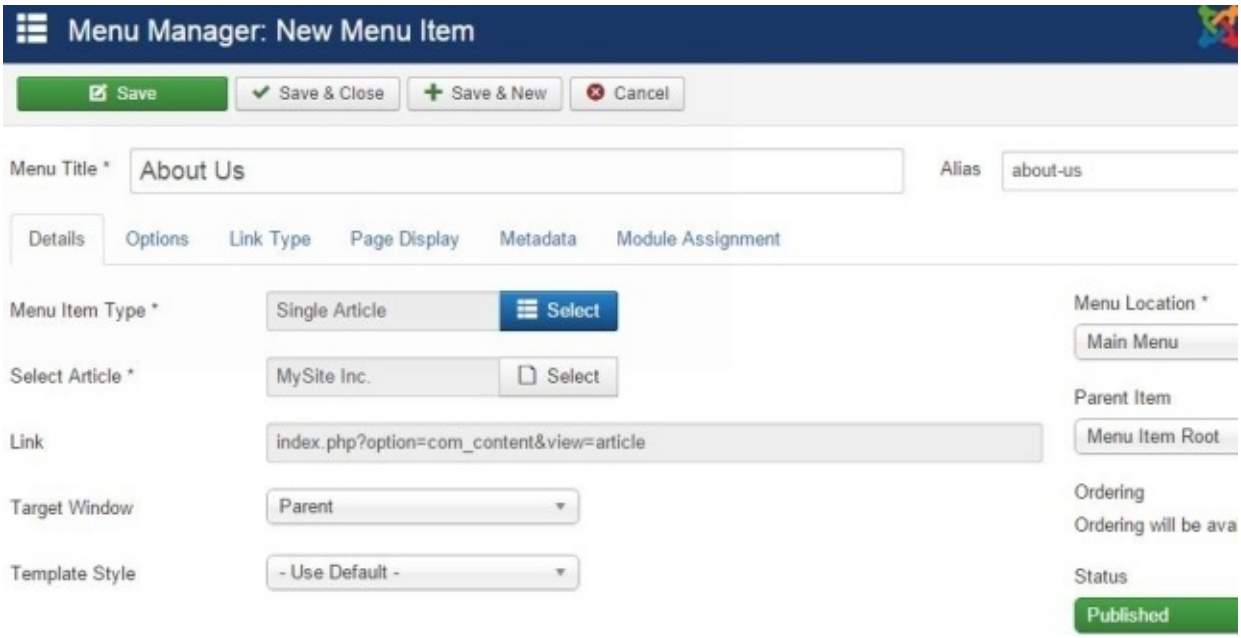
Joomla by default, adds a Main Menu under the Menu option, with a Home menu item in it, which actually enables us to view the default homepage after the Joomla installation. So we start by adding a new menu item to appear after Home on the pages of the website.

We navigate to Menus Main Menu Add New Menu Item.

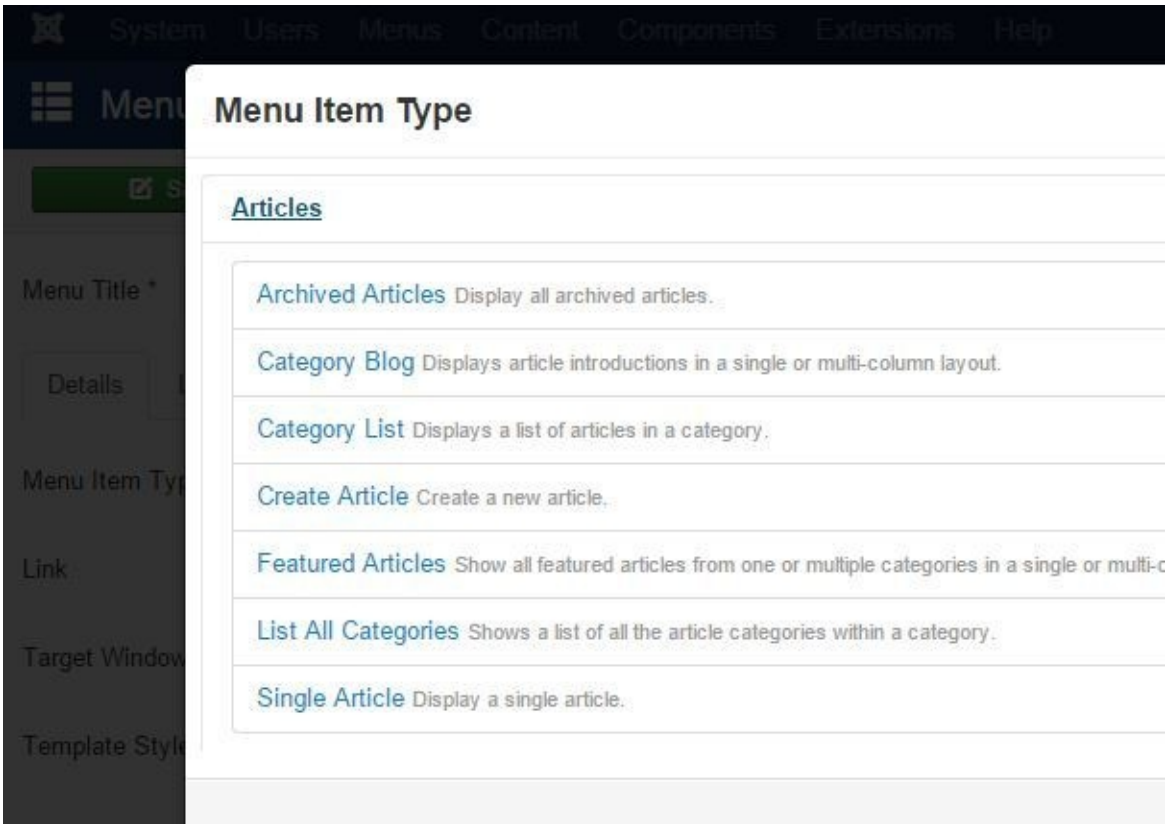


Now we want to create a page to display ‘About Us’. So we name the menu item as ‘About Us’ and set the alias as ‘about-us’. This defines the path of the menu item. If left empty, Joomla automatically inserts the date and article id in the path. But that looks bad on the address bar and is also unfriendly for the search engines.

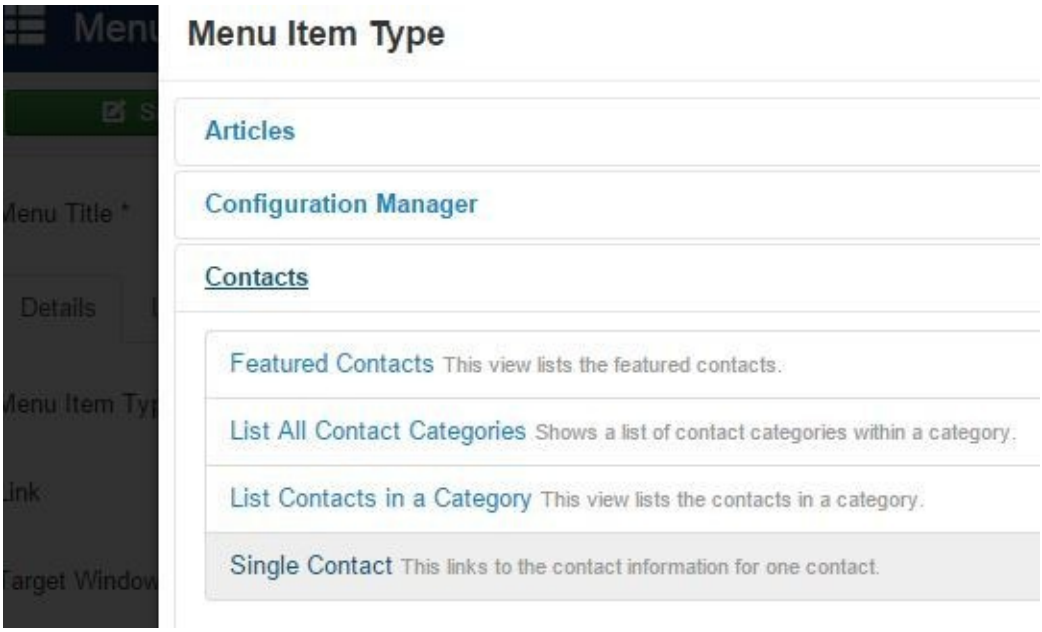
Moving forward, we set the Menu Item Type as Single Article from ‘Article’, then select the article to be displayed when someone clicks on About Us. So we select ‘MySite Inc.’ – the article which we created previously to serve as an information page about the business.



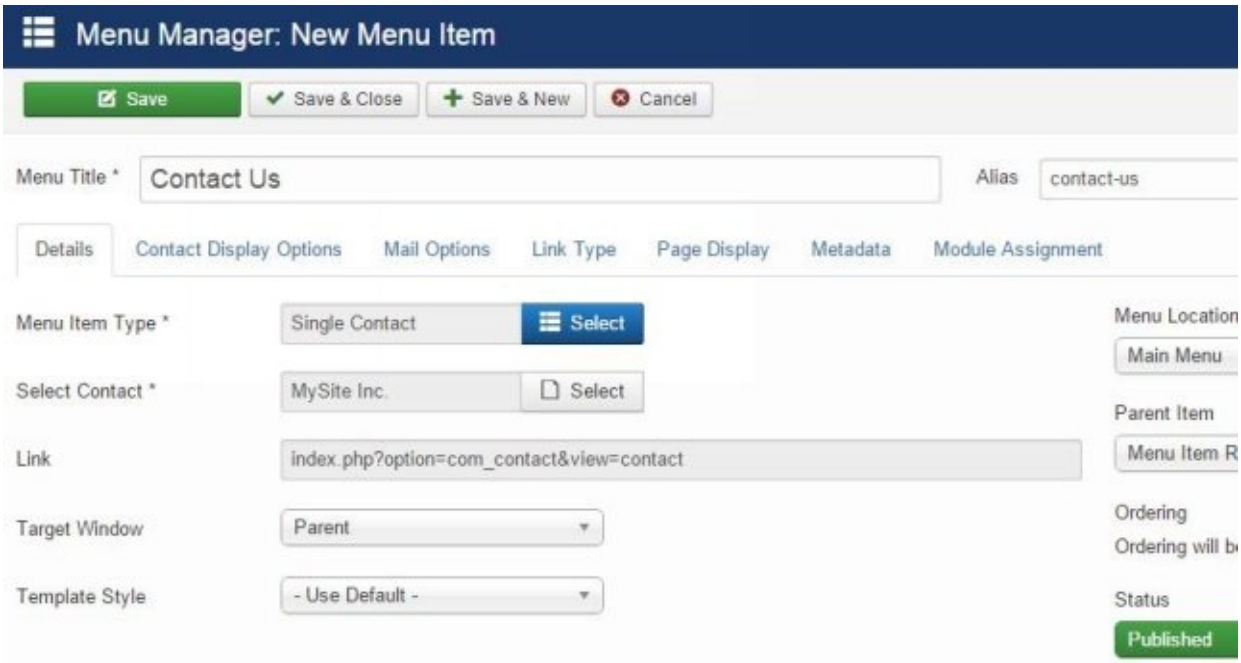
We repeat the same method to add another Menu Item called ‘Services’ to display the article that shows up the service list of the business. We use the same way to select the Menu Item Type followed by Selecting the Article.



Finally we need to add another Menu Item to display the contact information along with a contact form. To do this, we click on New in the ‘Menu Manager: Menu Items’ page and select Single Contact as the Menu Item Type, as shown below:



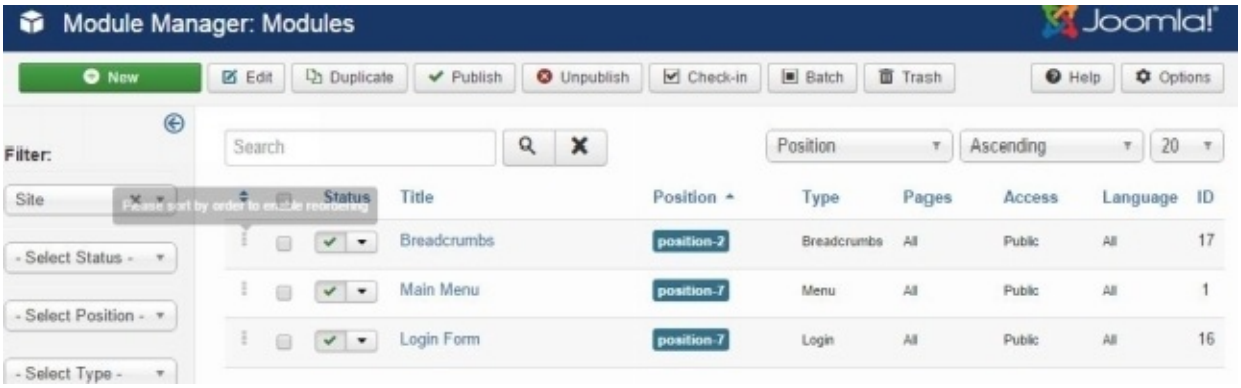
This determines that this menu item will be displayed from the Contacts component and not the Article or Content component. We choose the contact MySite Inc. from the list and Save & Close the menu item.



3.2.5 Displaying the Menu

The final stage is to display the menu we just created. To display the menu items, we need a module to be assigned on the website's frontend. To do this, we go to Extensions Module Manager.

Joomla installation displays the main menu through a menu module by default. The module is set to display at **position-7** of the default template. See Quick Tips in Section 3.3.3 of this book to learn about assigning Modules to certain positions.

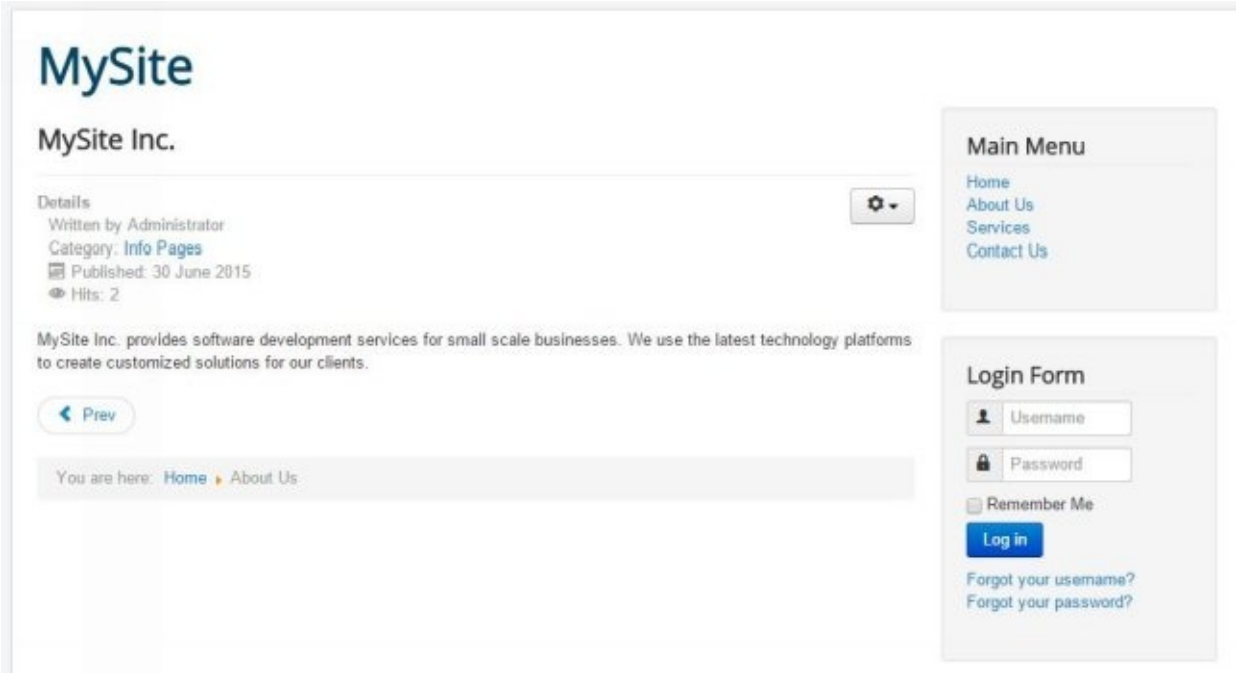


So, apparently, our work is done. Now we will check the frontend of the website to see the result of what we have been doing so far.

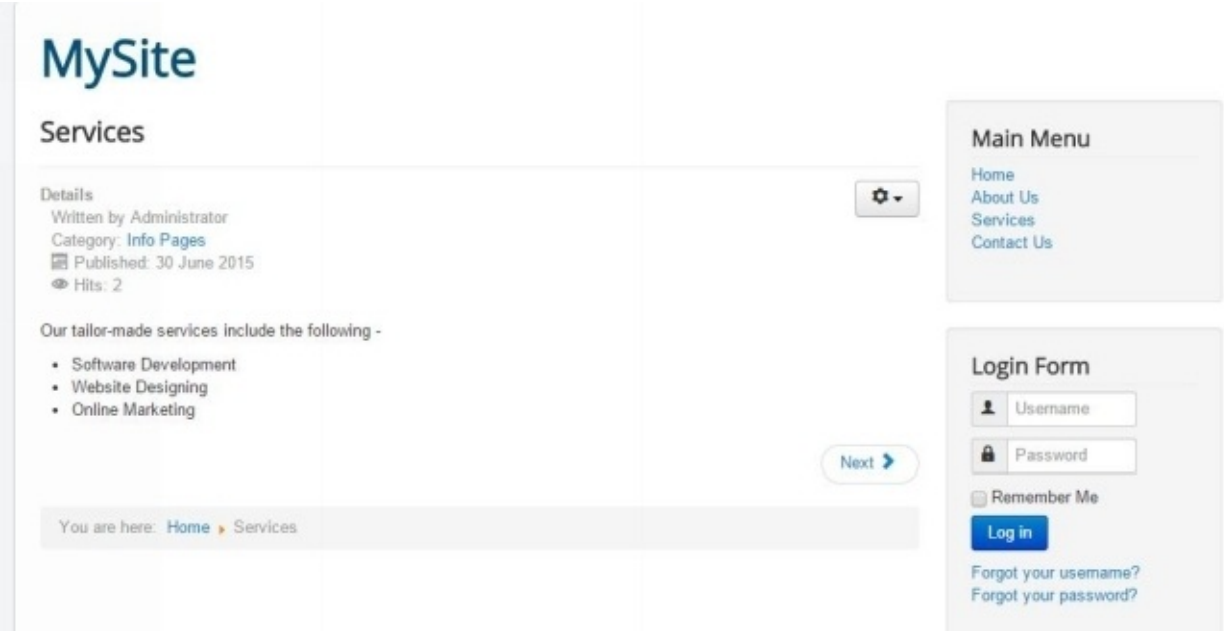
Homepage:

We can see that the Menu Items are visible on the right side of the page. Now we will click on every link on that menu and check the pages.

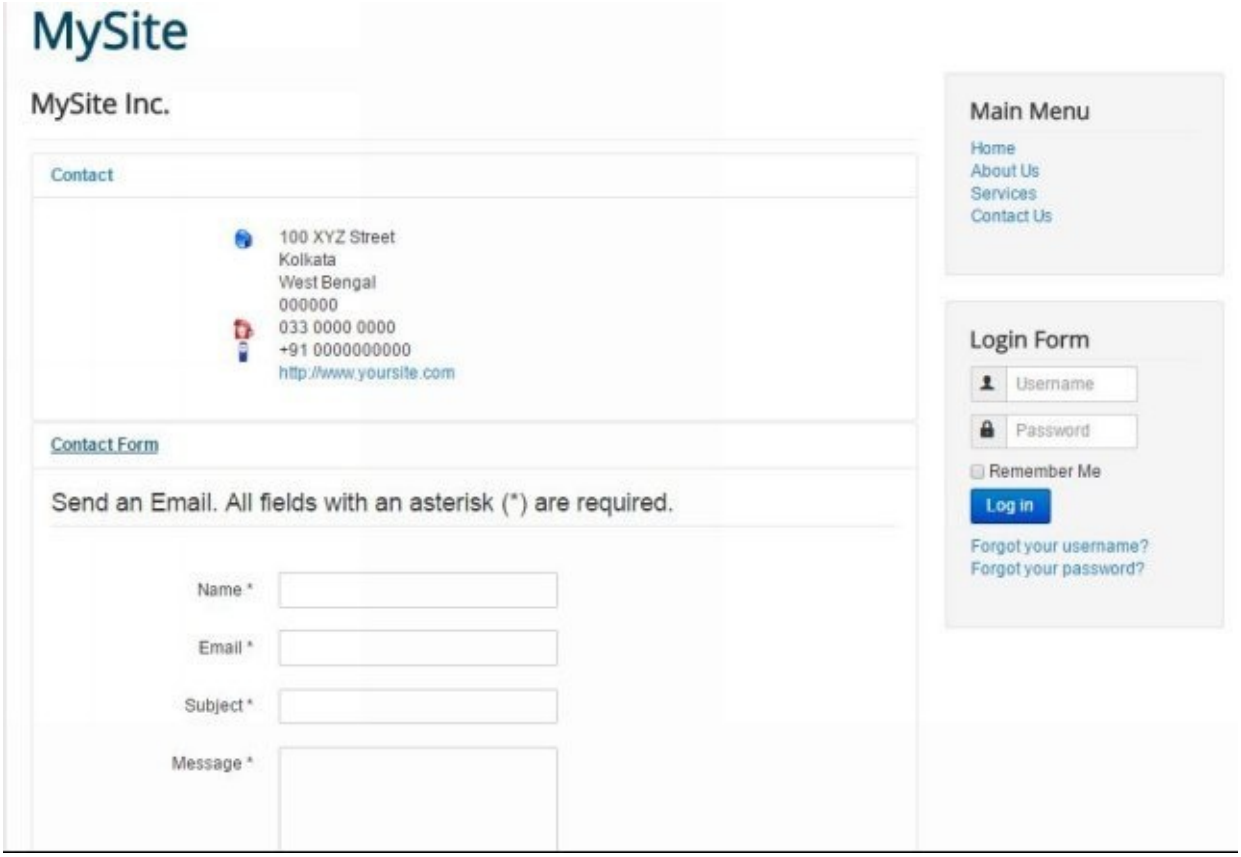
About Us:



Services:

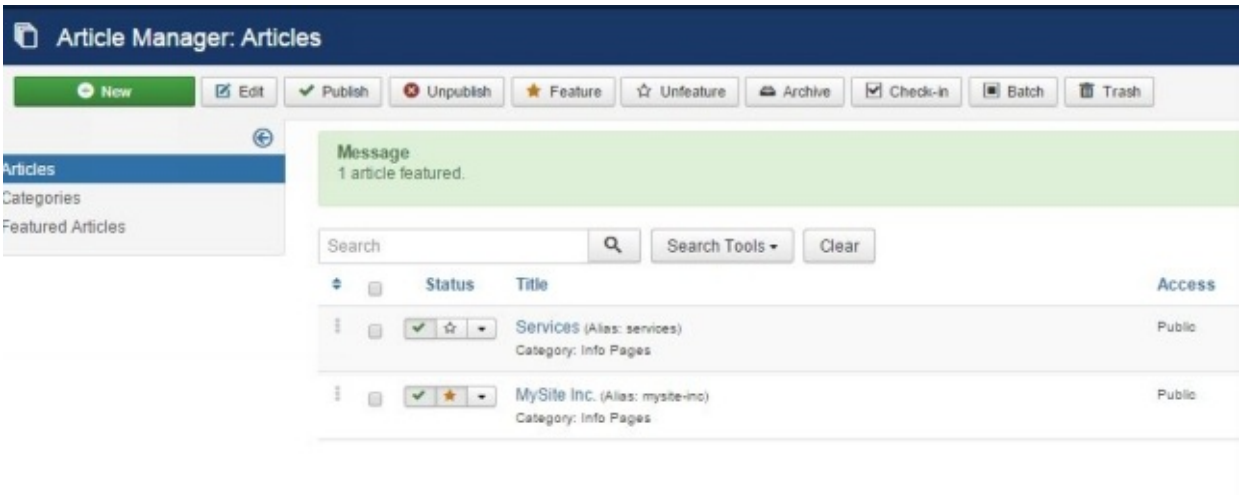


Contact Us:

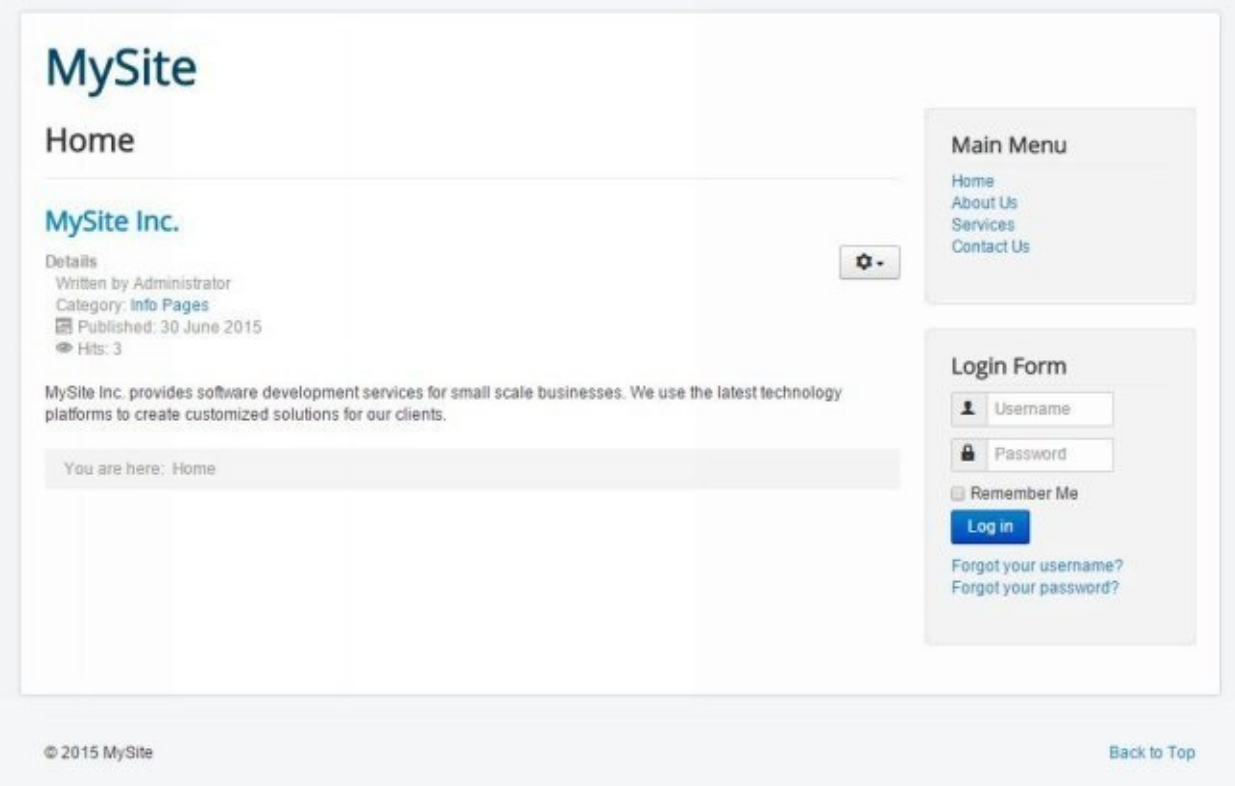


So we have successfully created a very basic website. Now if we want to display the About Us article on the Homepage, we can do it by simply featuring the Article from the Article Manager. This is because, we have learnt previously that the default Home menu items displays Featured Articles from any category.

Featuring the article by clicking on the star button:



Checking the Homepage once more, for the update:



As we can now see, the article about the business is displayed on the home page of the website – MySite.

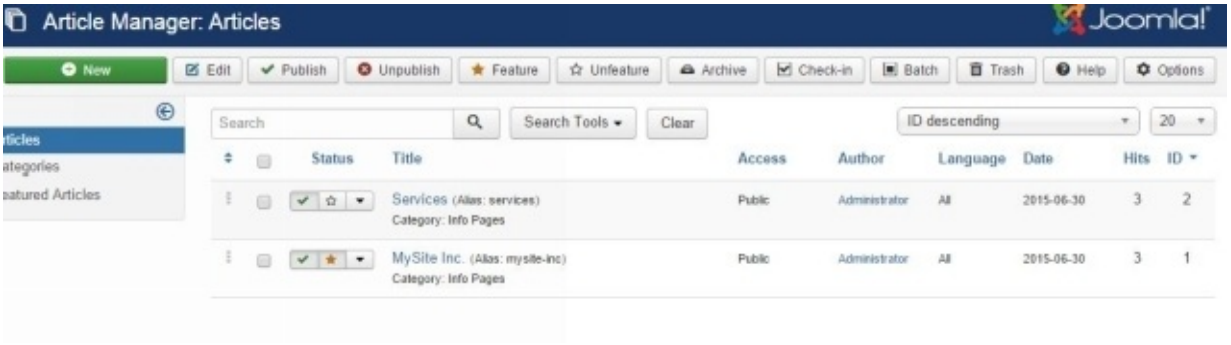
Now you can experiment by adding graphics, text content and other elements to the website to improve the visual appeal of the website. Our purpose of understanding Joomla CMS and setting up a website with it, has been served with this basic business website.

3.3 Tuning the Basic Settings

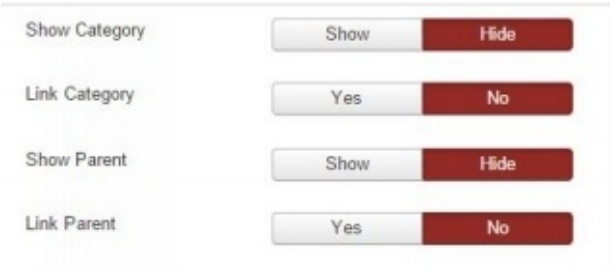
3.3.1 Article Settings

As we can see in the previous screenshots, every page displays a section of ‘details’ about the original article including the category name, author name, published date & number of hits. In a business website, these information are not required, as far as we are not writing a daily blog. If the website had a news section for company news updates, we could have let this info section to remain like that, as news articles have no value without a date and author details. So, to turn off this extra segment, we have to tune the article settings.

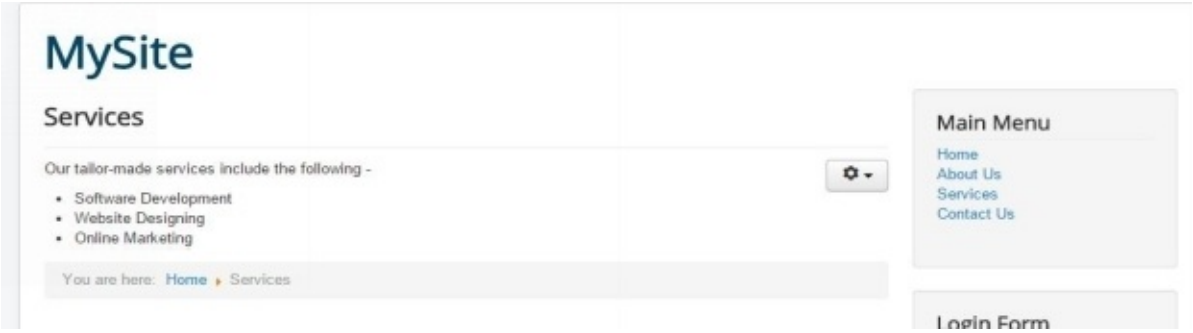
To do this, we navigate to Content Article Manager. Then click on **Options** button.



We turn the non-required fields off as shown below & click on the Save button.



Now we shall check the *Services* page for instance, on the frontend & see if it has worked or not:



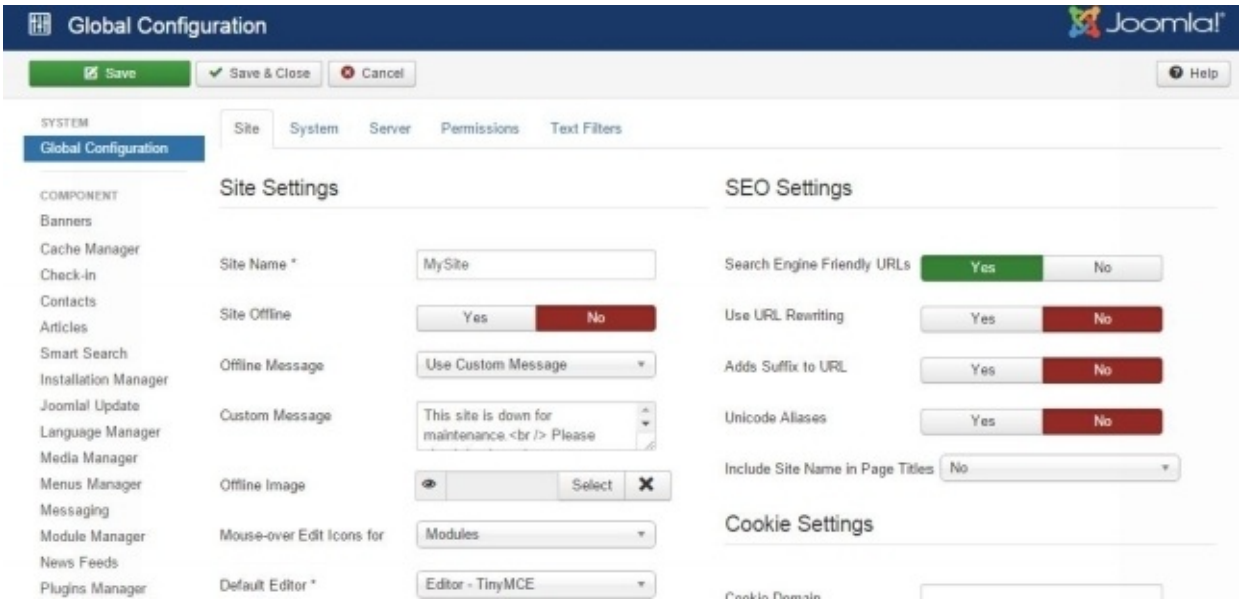
Now there’s no more extra detail about the article & this looks more professional.

3.3.2 Site Settings

Although we should have navigated to this section before the Article Settings, I chose to come here later, because, most of the basic settings which appear under this section has been pre-defined by us, while installing Joomla in the first place. Remember the Joomla

Installation page? Well, this is the same data as far as basic settings are concerned, like the site name, description & database settings. However, there's a lot more we can achieve through this Site Settings page.

We first navigate to System Global Configuration



Remember, every time you click on **Options** inside a **component**, like we did under the **Article Manager**, you actually open the system settings of that particular component, which is actually located here. You can see that on the left side of the image above (*bordered with red*).

On the other side of the page, we see a lot of options, including the option to put your **site offline**, until you are done making it, so that visitors cannot view the under construction / incomplete site. Currently this setting is set to 'No' (*bordered with green*) as we are making the site on a local PC and there's no scope for a public visitor.

A default Joomla installation comes with two text editors – **TinyMCE** & **CodeMirror**. The first one gives us an easy-to-use interface with all the basic text editing needs like <bold>, <italic>, <underline>, etc. a button away. But when working with the raw HTML, it is always recommended to use the CodeMirror text editor, which gives full control to the developer. So we would set the Default Editor to **Editor - CodeMirror**.

Now you can write your articles & create HTML modules in Joomla in HTML, rather than using a WYSIWYG text editor. However, please keep in mind that you need not use the <head> & <body> tags inside an article or a module. You can directly start with <p> tags. Joomla does the rest for you.

Coming back to the Global Configuration, there are a number of other settings that you may find useful under the System, Server, Permissions or Text Filters tabs. Now that we know about the basics, I would suggest you to experiment and learn about these options by yourself. The more mistakes you commit, the more you learn.

3.3.3 Quick Tips

To view all module positionsavailable in a template, go to Extensions Template Manager. Click on **Options** and switch '*Preview Module Positions*' to '**Enabled**'.

Then open the frontend of your site & add this to the URL: "/?tp=1"

For example: *http://www.yoursite.com/index.php/?tp=1*

To turn on New User Registration fromthe frontend of your website, go to Users User Manager and click on Options to retrieve the Global Configuration related to users.

Then simply switch '*Allow User Registration*' to '**Yes**'. You can also configure other settings available on the same page & click Save & Close to reflect the changes on your site.

Sending out Newsletters to the registered users on your site is really easy. Navigate to Users User Manager Mass Mail Users.

Then select the User Group who you want to receive your email. Fill up the text areas with your newsletter content & click on **Send email**.

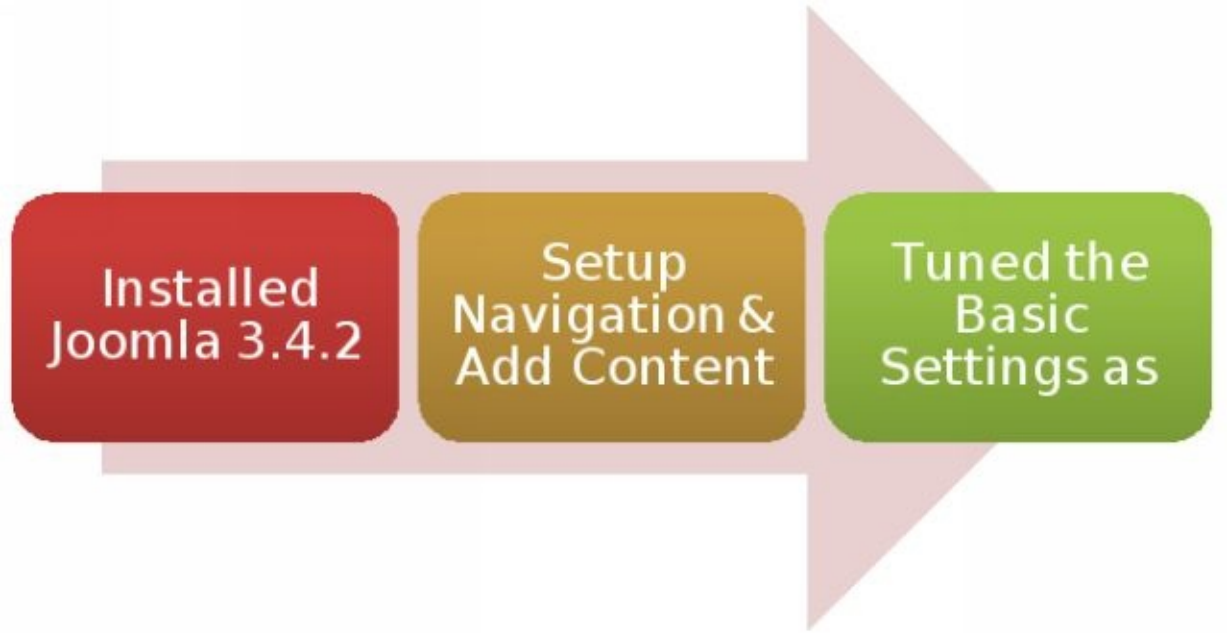
Display a Module on any available position of the site. Simply go to Extensions Module Manager. Click on **New** & select the type of module you want to display from the list. For example, let's choose **Articles – Latest.** Fill up the Title field, choose from the various settings to customize the module. Remember to select a **Position** from the right panel. Don't forget to assign the menu from **Menu Assignment.** From here you can choose which web pages shall display your new module. After it's done, click Save & refresh the frontend to check out the module.

Display a Module inside an article. Simply assign that module to an undefined position for example: *position-xyz***.** Then type **{loadposition position-xyz}** in the article, wherever you wish the module to appear.

3.4 Conclusion

We successfully installed a Joomla 3.4.2 website and have also created a very basic website with the following navigation system:

We have executed the following steps in achieving this –



Now you know, what Joomla! CMS is, and how to set up a basic website with it.

There is more to Joomla! as a Content Management System. Also to set up a Joomla website is not enough to start a fruitful business. Setting up a website is the first step towards a whole new world of possibilities. With proper design and targeted promotion, a website becomes a brand. We will learn about preparing complex web solutions using Joomla in our next book, which would focus on the applications of Joomla!

Beginning with Joomla! CMS

By Utsav Bhanja

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